novam-research.com - About entirely novel and environmentally friendly energy technologies and other topics, concepts, and results of non-mainstream science & Funding call to advance them

Last updates Aug 2025 - 7 Mar 2026 on this home page & here1 here2 here3 here4 here5 here6 here7 here8

Welcome to this website which is aiming to contribute to a progress and paradigm shift in physics, science, technology, awareness, and to an enjoyable evolution of mankind.

Dr. Frank Lichtenberg
Physicist

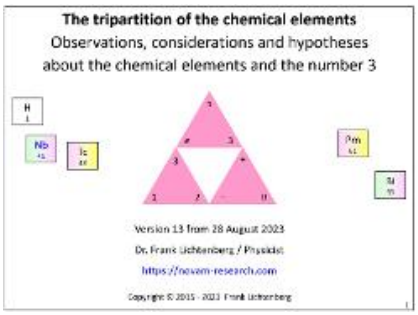
Topics on this home page further down:

- Entirely novel and environmentally friendly energy technologies, i.e. various self-running and zero emission motors / generators / power plants which tap an everywhere present form of energy which is called space energy, vacuum energy, zero-point energy, or ether energy
- New entry since 27 September 2025: Industrial size 1 MW generators from the company GEM Energy / Sinag - modularly scalable up to 400 MW
- Generators based on autonomous undamped hydraulic oscillations
- ECAT technology / generators
- Magnet motor / generator systems from the Quantum Energy (Inductance Energy) Corporation
- Solid state power amplifiers from the company Holcomb Energy Systems
- Patent DE3501076A1 from 1986 by the former Siemens engineer Dr. Wolfgang Volkrodt

Presentation about entirely novel and environmentally friendly energy technologies. Version 100 from 7 February 2026, 4 MB pdf

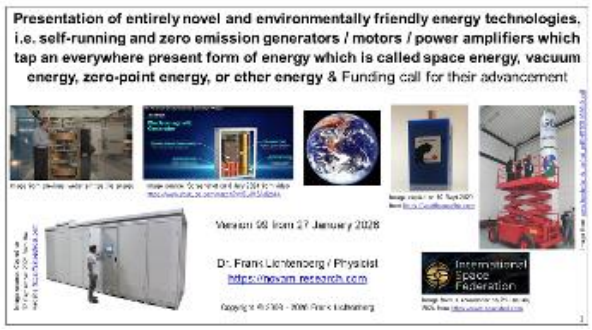
Presentation of a research project: Searching for Room Temperature Superconductors. Version 51 from 22 October 2023, 11 MB pdf

- Solid state generators from the SEMP Research Institute Co. Ltd.
- Various types of magnet motors & New German-language book "Autonome Magnetmotoren" (2024)
- Cryogenic magnet motor of Walter Thurner
- Tewari generator
- So-called neutrino voltaic
- Reid Cell & New book "Reid Cell" (2024)
- Kinetic power plants from the company Rosch Innovations
- Water splitting by pulsed electrolysis
- Alpha Institute of Advanced Studies (AIAS) - ECE Theory
- International Space Federation & Their significant paper "The Origin of Mass and the Nature of Gravity" by N. Haramein, C. Guermonprez, and O. Alirol
- Tom Bearden: His website, motionless electromagnetic generator (MEG), and book "Energy from the Vacuum"
- Examples of papers including abstract about physical concepts for entirely novel energy and propulsion technologies
- Superconductivity
- The tripartition of the chemical elements
- Synthesis and study of special oxides
- Harmful effects of electrosmog and wireless communication technologies



Presentation about the tripartition of the chemical elements. Version 13 from 28 August 2023, 1 MB pdf

Entirely novel and environmentally friendly energy technologies



Presentation about entirely novel and environmentally friendly energy technologies, i.e. self running and zero emission generators / motors / power amplifiers which tap an everywhere present form of energy which is called space energy, vacuum energy, zero point energy, or ether energy & Funding call for their advancement. 95 pages / slides, 4 MB pdf, version 100 from 7 February 2026

New entry since 27 September 2025

Entirely novel and environmentally friendly energy technologies

Self-running and zero emission industrial size 1 MW generators from the company GEM Energy / Sinag - modularly scalable up to 400 MW

Creation date 27 September 2025

Last update 7 February 2026

A group of people in front of a 1 MW generator EER-SPG. Four of the seven front doors of the generator are open. Image source; https://sinagglobal.com . Picture copied from that website on 23 November 2025 and uploaded here

The inventor Bonifacio J. Eyales and the company Sinag / GEM Energy have developed self-running and zero emission industrial size generators which are called EER-SPG. EER stands for Electromagnetic Energy-flux Reactor and SPG means Stationary Power Generator. It is patented in many countries, see for example the United States patent US9444264B2 . The corresponding patent application was already published in 2013 and the granted application was published in 2016.

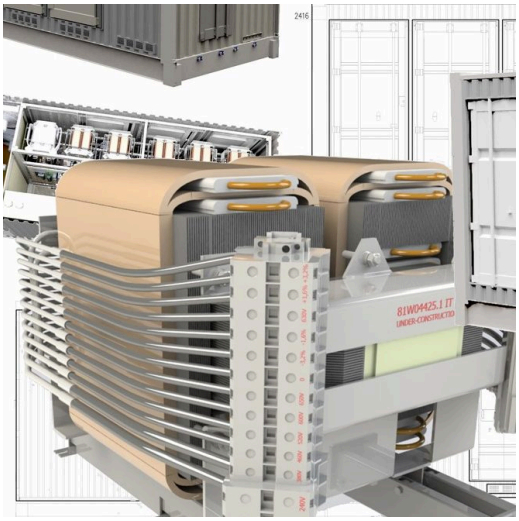


Image source: https://sinag.energy - Picture copied from that prior website on 27 September 2025 and uploaded here

Website of the German company GEM Energy Deutschland / Sinag: https://gem-energy.de (prior https:/sinag.energy) . The German company GEM Energy Deutschland / Sinag offers 1 MW generators (400 V AC, 50 Hz or 480 V AC, 60 Hz). They are modularly scalable up to 400 MW. An English-language technical data sheet of the 1 MW generator can be downloaded from this website (as of July 2025): https://novam-research.com/resources/Sinag-1MW-generator-EER-SPG_Technical-data-sheet_July-2025.pdf . Therein the specified dimensions and weight of an EER-SPG are 6,8 x 2,8 x 1,2 m and 12600 kg. However, the series model has a new design with a smaller set-up area and a smaller weight, namely smaller than 8 square meter and about 9000 kg.

Picture of the outer appearance of a 1 MW generator EER-SPG. Image source https://sinagglobal.com . Picture copied from that website on 27 September 2025 and uploaded here

The production of the 1 MW generators EER-SPG has already started at several locations in Europe and the delivery time is currently 8 months (as of January 2026) and is expected to be 6 months when the series production is fully established. After an intended launch event in Germany - sometime after the first quarter of 2026 - an operating 1 MW generator can be publicly viewed at an industrial company in Germany. The company GEM Energy Deutschland / Sinag is currently also contact partner for the distribution and installation of the generators EER-SPG in Europe.

Picture of the outer appearance of the new design of the series model of a 1 MW generator EER-SPG. Image source: Screenshot from https://gem-energy.de on 7 February 2026

The generator requires once per year a maintenance which will be performed by staff members from the corresponding national company GEM Energy. Mainly the liquid cooling system and the ventilators, which are the only moving parts in the generator, need such a periodic maintenance.

The 1 MW generator(s) cannot be purchased but will be provided at the customer location by a leasing-like purchase agreement and the generated and consumed electricity is paid by using a specific tariff or rate. Furthermore, the generators are supposed to feed electricity into a microgrid or local grid, e.g. into that of an industrial company, a village, or a city. The generators shall not be used to feed electricity into the public electricity grid, i.e. they are supposed to support a decentralized energy supply instead of an expansion of the public electricity grid.

The manufacturing of the generators is financed by joint venture partners. Currently there are already joint venture partners for the countries Philippines, Germany, Malaysia, Japan, Australia, and California in USA (as of December 2025). The certification of the generators takes place by the global certtification company Det Norske Veritas (DNV).

Websites of the company GEM Energy Philippines and Sinag Global in the Philippines: https://sinagglobal.com and https://gempowerph.com

YouTube Video of a generator EER-SPG, visible is only its outer appearance and hearable is its noise: https://www.youtube.com/watch?v=aWqFL779kEw&t=1s

Webpage of the Austrian Society for Space Energy (ÖVR) about the generator EER-SPG: https://oevr.at/philippinischer-elektromagnetischer-energieflux-reaktor-stationaerer-energie-generator-eer-spg-1-mw-anlage

On 11 October 2025 Alex Klass from the company GEM Energy Deutschland / Sinag Deutschland presented a German-language talk at a congress about novel and environmentally friendly energy technologies and related topics in Germany. A German-language report about the congress, which took place on 11 and 12 October 2025, is published on pages 4 - 21 in the November / December 2025 issue of the German-language journal NET-Journal (ISSN 1420-9292). It reports on pages 6, 7, 9, and 12 about the presentation of Alex Klass on 11 October 2025 which included also a live video hook-up to the inventor Bonifacio J. Eyales from the company GEM Energy / Sinag in the Philippines (he did speak in English) and a questions and answers session.

On 13 December 2025 a German-language congress about novel and environmentally friendly energy technologies and related topics took place in Zurich in Switzerland. It was organized by Adolf and Inge Schneider and there they showed a video-like presentation (slides plus voice) from Alex Klass about the EER-SPG. A German-language report about that video-like presentation is published on pages 6 and 7 in the January / February 2026 issue of the German-language journal NET-Journal (ISSN 1420-9292).

About the operating principle of the generators EER-SPG

Let's first comment the term "reactor" which is part of the name of the generator and is used in the above-mentioned patent, technical data sheet, and websites. The term "reactor" could be confusing because one could associate that terrm with a device which contains a plasma, gas, or liquid. However, here the term "reactor" means a specific electromagnetic component or setup which consists mainly of coils and a ferromagnetic material which is magnetically soft, e.g. something which is similar to a transformer.

According to the above-mentioned websites and the technical data sheet the EER-SPG is based, among others, on a specific design which eliminates the Back

Electromotive Force (Back EMF) and uses as energy source ions and electrons from the atmosphere. In the EER-SPG the induction of a voltage and current in a conductor by a magnetic field takes place in a setup which comprises coils and a ferromagnetic material which is magnetically soft such as a transformer-like construction.

The view from the author of this website is the following:

Macroscopic consideration:

- A design which allows the elemination of the Back EMF leads to the creation of a system which generates an output energy or power that is larger than its input energy or power. This in turn allows the construction of a self-running generator by feeding a part of its output energy or power into its input or by making sure that the output voltage keeps within certain limits. Electrons and ions from the atmosphere do not play a role in this.

Microscopic perspective (general description without knowing the involved detailed processes which also depend on the type or construction of the generator):

- All material manifestations including man-made devices such as a generator consist of atoms and their constituents, namely electrons, protons, and neutrons. The atoms consists more than 99,9 % of vacuum or so-called empty space - which is actually not empty - because the atomic nucleus (protons and neutrons) is tiny compared to the atomic shell (electrons). The source of the usable output energy or power of any type of self-running generator are not electrons and ions from the atmosphere and also not the electrons and protons which are the constituents of the atoms of which the generator is made of. Instead, any type of self-running generator taps via the electrons (and / or protons) and their specific properties an everywhere present form of energy which is called space energy, vacuum energy, ether energy, or zero-point energy. The space, vacuum, or ether can be considered as an everywhere present (fluid-like) medium which comprises, among others, a huge amount of energy or energy density and the energy needs of humanity is tiny compared to that. Particles like electrons or protons, which are the constituents of atoms or ions, are special oscillation and / or vortex states of that medium. For references and further information see, for example, this information document and on this home page the section "Examples of papers including abstract about physical concepts for entirely novel and environmetally friendly energy and propulsion technologies"

Acknowledgement

The author of this website thanks

- Adolf and Inge Schneider for presenting some information about the company Sinag and their novel energy technology during a German-language meeting of

the Swiss Association for Space Energy (SVR) on 19 September 2025 in Zurich in Switzerland

- Adolf and Inge Schneider and Alex Klass from the German company GEM Energy Deutschland / Sinag Deutschland for several discussions

- Alex Klass from the German company GEM Energy Deutschland / Sinag Deutschland for providing on 2 October 2025 an English-language technical data sheet of the 1 MW generator (as of July 2025)

- Alex Klass from the German company GEM Energy Deutschland / Sinag Deutschland who presented on 11 October 2025 a German-language talk at a congress about novel energy technologies and related topics in Germany. A German-language report about the congress, which took place on 11 and 12 October 2025, is published on pages 4 - 21 in the November / December 2025 issue of the German-language journal NET-Journal (ISSN 1420-9292). It reports on pages 6, 7, 9, and 12 about the presentation of Alex Klass on 11 October 2025 which included also a live video hook-up to the inventor Bonifacio J. Eyales from the company GEM Energy / Sinag in the Philippines (he did speak in English) and a questions and answers session

- Adolf and Inge Schneider and their support team for organizing a German-language congress about novel and environmentally friendly energy technologies and related topics on 11 and 12 October 2025 in Germany.

  There Alex Klass from the German company GEM Energy Deutschland / Sinag Deutschland presented a talk on 11 October 2025, see previous item

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Entirely novel and environmentally friendly energy technologies

Self-running running and zero emission generators which are based on autonomous undamped hydraulic oscillations and the ram pump principle

Last update 15 February 2026

The content of this section of the home page is also available on a separate webpage

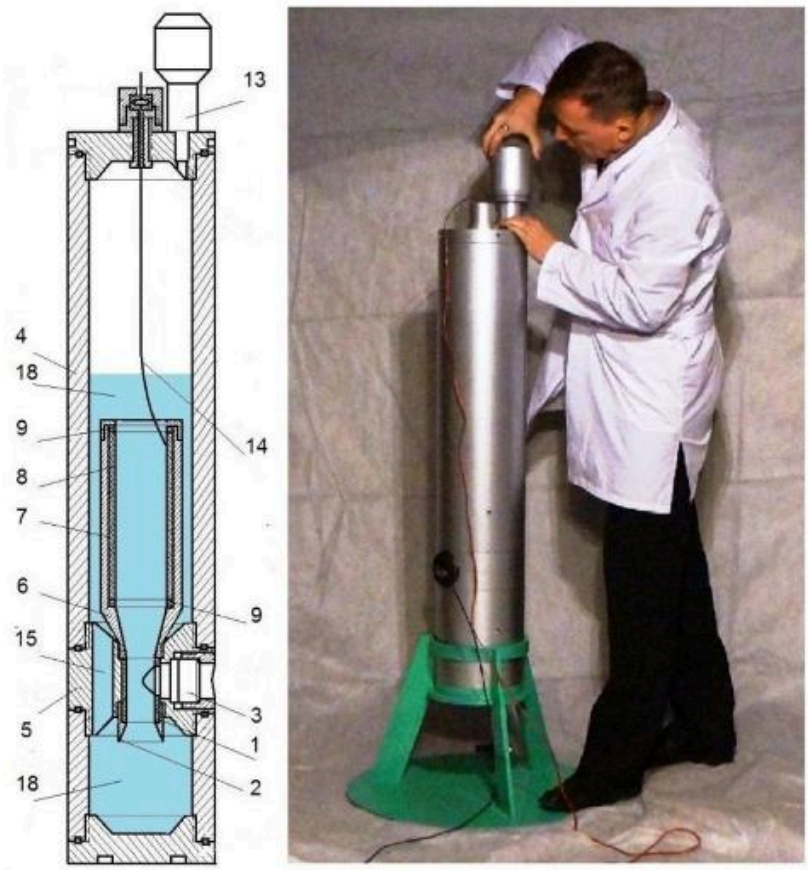


Image source: Figure 19 from a paper by Dr. V. Marukhin and Dr. V. Ivanov about ram pumps and self-running generators based on undamped hydraulic oscillations

Starting from the operating principle of so-called ram pumps Dr V. Marukhin and his co-workers have developed various types of self-running generators which are based on autonomous hydraulic oscillations. An example is shown in the picture and sketch on the left which were copied from Figure 19 of a paper by Dr. V. Marukhin and Dr. V. Ivanov.

In the sketch above the blue color represents a liquid such as water and the white space above is filled with a gas such as nitrogen. Both the liquid and the gas are put under a high pressure of up to 3000 bar depending on the design or type of the generator. To initiate the self-running (undamped) vertical fluid and gas oscillations it is necessary to apply a specific high pressure pulse for a short time. That is realized by an electrically ignited explosion of about 2 g gun powder. The resulting pressure wave is transferred into the generator via a shock valve. The frequency of the self-running (undamped) vertical fluid and gas oscillations can be up to 3000 Hz depending on the design or type of the generator. The self-running hydraulic oscillations are converted into a pulsed high DC voltage by piezoelctric transducers. The pulsed DC voltage can be up to 12 kV depending on the design or type of the generator. Via a DC / DC and DC / AC converter a convenient form of electricity can be obtained. The resulting usable output power can be up to 1 MW depending on the design or type of the generator. The following and already above-mentoned paper by Dr. V. Marukhin and Dr. V. Ivanov reports about a 125 kW generator which worked non-stop for 2 years, connected to different loads and at different ambient

temperatures: https://novam-research.com/resources/Generators-based-on-self-running-hydraulic-oscillations_Paper-by-V-Marukhin-and-V-Ivanov.pdf

Slide set (67 pages, 3 MB pdf, mainly in German) of a German-language talk by Adolf Schneider from October 2025 about ram pumps and self-running generators based on autonomous undamped hydraulic oscillations. His talk took place on 12 October 2025 in Germany at a German-language congress about novel energy technologies and related topics.

A German-language report about that congress is published on pages 4 - 21 in the November / December 2025 issue of the German-language journal NET-Journal (ISSN 1420-9292). It comprises on pages 18 and 19 the German-language sub-report "Revolutionärer hydraulisch-elektrischer Generator bis 1 MW" about the generators developed by Dr. V. Marukhin and his co-workers.

The following table from page 44 of the slide set from Adolf Schneider displays an overview of the self-running generators which are based on autonomous undamped hydraulic oscillations which were developed by Dr V. Marukhin and his co-workers (as of August 2025) (there and in the slide set from Adolf Schneider the generators are called HEG which stands for Hydraulic Energy Generator and "HEG-Typ" means HEG type):

HEG-Typ	year	power [kW]	height [m]	diameter [m]	weight [kg]
Module RAS	2015	500	1.00	0.20	
YPHP20SP-B	2015	20	0.20	0.04	0.9
YPHP100-C	2015	100	0.47	0.14	32
YPHP1500A	2015	1500	1.00	0.20	
APC-855-16	2016	890	0.50	0.20	95
H1000VS	2017	1000			
EGMP-H-250-18	2018	250			
EGMP-Air-250-19	2019	250			
EGMY-750-19	2019	765	0.48	0.14	37
EGM-H-P-500-20	2020	500	0.30	0.14	20
EGM-H-P-1000-21	2021	1000	0.42	0.15	40
EGMHY-750-22	2022	725	0.45	0.14	26
EGMYS-2000-25	2025	1540	0.45	0.15	38
EGMHS-2000-25	2025	2050	0.43	0.15	39
EGMHR-750	2025	750	0.43	0.09	14

The generator types EGMHS-2000-25 and EGMHR-750 can be delivered with an associated DC / DC and DC / AC converter which provide a stable output voltage of 3500 V DC and 220 V AC, respectively (as of August 2025).

The generators are not marketed publicly. **As of August 2025:** According to Dr. V. Marukhin there are 23 generators in operation at undisclosed industrial customers in the following countries: Spain, Portugal, Morocco, Saudi Arabia, United Arabic Emirates, Madagascar, China, and Russia. In Spain there is a holding company of Dr. V. Marukhin and his co-workers which comprises also a manufacturing site. There only shareholders of the holding company can order generators. The manufacturing of the generators takes also place in some of the other above-mentioned countries on behalf of the corresponding license partner. The latter is authorized to manufacture generators only for its own use. The production time in Spain is 1 month, in China 4 months, and in Russia 6 - 8 months.

It is intended to acquire for some countries a license for the most recent version of the 1 MW generators and make for Switzerland, Germany, and Austria some demonstration generators available which can be viewed and inspected on request. See also the following flyer from the Swiss company Overunity GmbH from 2023: http://www.borderlands.de/Links/Projekt%20HEG-1000kW-OU-010623-EN.pdf . However, the implementation of this intention is impeded because of the imposed

sanctions against Russia. Therefore the desired demonstration generators can be available earliest in 2026 or 2027.

About the operation principle of these self-running generators which are based on autonomous undamped hydraulic oscillations

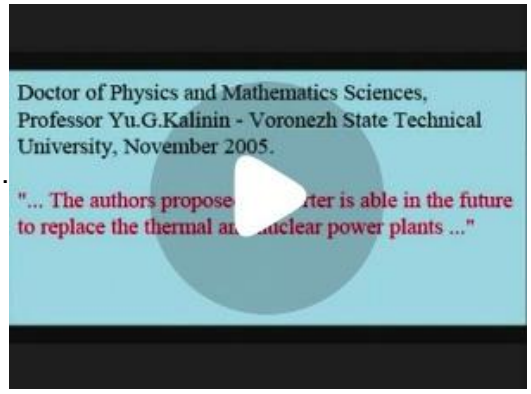
The russian inventors Dr. V. V. Marukhin and V. A. Koutienkov have developed a macroscopic theory which comprises about 500 formulas (fluid mechanics, fluid dynamics, hydrodynamics, hydraulics etc.). It allows the design of a new type of ram pumps without water losses. Under specific geometrical and dynamic conditions these systems work autonomously.

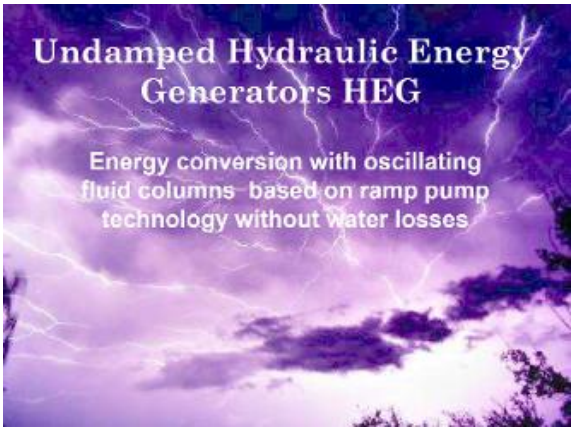
Microscopic perspective (general description without knowing the involved detailed processes): The author of this website assumes that this and any other self-running energy technology extracts usable energy from an everywhere present form of energy which is called space energy, vacuum energy, zero-point energy, or ether energy as described in this information document . The tapping of this form of energy happens probably via the earth`s gravitational field which acts on the water and gas molecules and / or via electromagnetic fields that act at the atomic level between the water and gas molecules and / or via the electromagnetic fields that act at the atomic level between the atoms of solid materials and the water and gas molecules.

Further references and information

Paper by Dr. V. Marukhin and Dr. V. Ivanov about ram pumps and self-running generators based on undamped hydraulic oscillations. It can be downloaded via borderlands.de or via this website . This paper reports, among others, about an electrical 125 kW generator which worked non-stop for 2 years, connected to different loads and at different ambient temperatures.

Video from 2005 about ram pumps and self-running generators based on undamped hydraulic oscillations: https://yadi.sk/d/9CN509gHhoWgq . Language in English and labeling of the sketches or pictures mainly in Russian





Presentation (40 pages, 3 MB pdf) from May 2021 by Adolf Schneider about ram pumps and generators based on self-running hydraulic oscillations. It can be downloaded also via http://www.borderlands.de/Links/HEG-Presentation-English.pdf

German-language slide set (55 pages, 3 MB pdf) of a talk by Adolf Schneider about ram pumps and generators based on self-running hydraulic oscillations on 4 October 2020 at a conference about novel energy technologies and related topics in Stuttgart in Germany.

Booklet or E-book "The Eureka Machine" by Adolf and Inge Schneider about ram pumps and self-running generators based on undamped hydraulic oscillations which were developed by Dr. V. Marukhin and his co-workers. 1st English edition from March 2018, A5 type format (15 cm x 21 cm) and 199 pages. Publisher Jupiter-Verlag in Switzerland. See https://jupiter-verlag.ch/start_n.htm (in German) --> Click on

"Ausgewählte Titel" and scroll down until this booklet or E-book appears. Direct link to a detailed description in English about the booklet: https://jupiter-verlag.ch/shop/detail_neu.php?artikel=133&fromMain=j . Or see also, for example, https://www.goodreads.com/book/show/4087the-eureka-machine . Direct link to a detailed description in English about the E-Book: https://jupiter-verlag.ch/shop/detail_neu.php?artikel=134&fromMain=j

German-language booklet "Die Heureka-Maschine" by Adolf and Inge Schneider about ram pumps and self-running generators based on undamped hydraulic oscillations which were developed by Dr. V. Marukhin and his co-workers. A5 type format (15 cm x 21 cm) and 199 pages. Publisher Jupiter-Verlag in Switzerland. See https://jupiter-verlag.ch/start_n.htm (in German) --> Click on "Ausgewählte Titel" and scroll down until this booklet appears. Direct link to a detailed description in German about this German-language booklet: https://jupiter-verlag.ch/shop/detail_neu.php?artikel=132&fromMain=j . Meanwhile 6th edition from March 2019

German-language webpage of the Austrian Associatian for Space Energy (ÖVR) about the generators from Dr. V. Marukhin and his co-workers (in German): https://oevr.at/heureka-maschine

About ramp pumps, a water-lifting device, and a self-running generator by V. Kutienkov and V. Marukhin - text and images from RexResearch and Adolf Schneider (in English apart from page 1 which is partly in German): https://novam-research.com/resources/Ram-pump-and-water-lifting-device-and-self-running-

generator_V-Kutienkov-and-V-Marukhin_Text-and-images-from-RexResearch-and-Adolf-Schneider.pdf

German-language report on pages 15 - 17 in the January / February 2022 issue of the German-language journal NET-Journal (ISSN 1420-9292)

German-language report on pages 14 and 15 in the November / December 2021 issue of the German-language journal NET-Journal (ISSN 1420-9292)

German-language report on page 30 in the May / June 2021 issue of the German-language journal NET-Journal (ISSN 1420-9292)

German-language report on pages 20 - 24 in the March / April 2021 issue of the German-language journal NET-Journal (ISSN 1420-9292)

German-language report on pages 18 and 19 in the January / February 2021 issue of the German-language journal NET-Journal (ISSN 1420-9292)

German-language report on page 18 in the January / February 2020 issue of the German-language journal NET-Journal (ISSN 1420-9292)

Paper about a hydraulic ram pump from Adolf Schneider (in German):
http://www.borderlands.de/Links/BeschreibungResonanzwidderAS051115.pdf

Acknowledgement

The author of this website thanks Adolf and Inge Schneider for various reports, presentations, and communications about this type of novel energy technology.

Entirely novel and environmentally friendly energy technologies

Self-running electricity generators based on the so-called ECAT technology

Last update 14 February 2026

Report about the ECAT technology

Image from https://ecatthenewfire.com

Summary (short version): Websites: Official ECAT website https://ecatthenewfire.com and https://e-catworld.com . The image on the left from the official ECAT website displays a small and self-running electricity generator which is called E-Cat SKL NGU 10 W. Some of its specifications: DC output power / voltage 10 W / 12 V, diameter 6 cm, height 3 cm, weight 100 g, and expected operational lifespan more than 11 years. This and also generators with the following output power can be pre-ordered: 100 W, 1 kW, 2 kW, 3 kW, 5 kW, 10 kW, 1 MW, and custom size. The 5 kW generator, for example, has the dimensions 15 x 25 x 50 cm and a weight of 6 kg. Datasheet of the generators: https://ecatthenewfire.com/wp-content/uploads/2025/01/E-Cat-Power-NGU-DataSheet-2024-12-3.pdf . The production of 1 MW generators has started in 2025 and the delivery of 1 MW generators for industrial customers has begun in December 2025 / January 2026. Some of the essential components of the ECAT electricity generators are a (small) vacuum tube which is filled with a low pressure inert gas, a magnet which influences the motion and spatial arrangement of electrons in the vacuum tube, and a special electronic control. It allowst the extraction of usable energy from an everywhere present form of energy which is called zero-point energy, vacuum energy, space energy, or ether energy. For details concerning the operationg principle see e.g.

- the paper E-Cat SK and long-range particle interactions (2019 and last update in October 2022) by Dr. Andrea Rossi
- the Japanese patent application JP2022188336A (abstract also in English) by the inventor Dr. Andrea Rossi which was published on 21 December 2022. The following pdf document provides parts of

the Japanese patent application JP2022188336A which are translated into German (pages 1 - 8 as of 15 Feb 2025) and other parts which are translated into English (pages 8 - 13 as of 15 Feb 2025): Click here (1) to download it from borderlands.de or click here (2) to download it from this website

- a German-language article Energiegewinnung über Plasma-Entladung by Adolf Schneider , published on pages 4 - 10 in the July / August 2025 issue of the German-language journal NET-Journal (ISSN 1420-9292)
- the considerations On Zero-Point Energy and the E-Cat by the Swedish research company Sagacitas AB

Entirely novel and environmentally friendly energy technologies

Self-running and zero emission magnet motor / generator systems from the company Quantum Energy (Inductance Energy) Corporation

Last update 11 August 2024

Dennis Danzik and co-workers from the former company Inductance Energy Corporation have developed a magnet motor, called Earth Engine, with a mechanical output power up to 25+ kW which can propel a mechanical device or an alternator with a maximal electrical output power in the range from 7 to 25 kW.

Image taken on 6 October 2019 from a video in the previous website https://ie.energy

The magnet motors of the Inductance Energy Corporation comprise a flywheel, a special asymmetric magnet array, a special triggering, and a control unit. To control the magnet motor an electrical input power of about 300 W is required. If the magnet motor propels an alternator, then these about 300 W can be provided by the

alternator. That results in an overall self-running and zero emission magnet motor / generator system with a maximal electrical output power in the range from 7 to 25 kW.

In the United States the Earth Engine is already in production on a small scale and in use at customers via leasing. The Inductance Energy Corporation has developed, manufactured, and installed 7 to 25 kW engines, capable of driving up to 4000 pounds (1814 kg) of inertia and delivering in excess up to 25 kW electrical power.

In August 2019 Inge Schneider and Adolf Schneider and others have visited the company Inductance Energy Corporation in the United States. Running prototypes were inspected at two different locations, about 15 different devices in various stages of development were viewed, and several discussions with various people including the inventor Dennis Danzik took place. In September 2019 the author of this website attended a meeting in Switzerland where it was reported about it. Information about the visit in United States are presented on pages 25 - 53 in a German-language slide set (61 pages, 5 MB pdf) from a talk by Adolf Schneider about magnet motors on 3 October 2020 at a conference about novel energy technologies and related topics in Stuttgart in Germany.

Meanwhile the activities of the Inductance Energy Corporation are transferred to the company Quantum Energy Corporation. In their website https://qree.energy there is under Magnet a webpage called Magnetic Propulsion, https://qree.energy/magnetic_propulsion.html , which is still under construction. The former website https://ie.energy of the Inductance Energy Corporation is still existing but displays no content. The magnet motors were called Earth Engine and are meanwhile called Photon Engine or photon-driven magnetic propulsion engine. The author of this website would prefer another name without the term photon because the term photon appears here somewhat inappropriate and somewhat misleading.

YouTube videos with Dennis Danzik from the Quantum Energy Corporation about the magnet motor and self-running magnet motor / generator systems:

- https://www.youtube.com/watch?v=qsGvt-RRjuk from 14 July 2023
- https://www.youtube.com/watch?v=VRvJjQ2F68M from 21 July 2023
- https://www.youtube.com/watch?v=xur5GAQFUlA from 4 August 2023
- https://www.youtube.com/watch?v=4vIuMapXN8I from 24 August 2023

- https://www.youtube.com/watch?v=lvZTomsskJo from 13 September 2023

Related video from 26 October 2023: https://www.youtube.com/watch?v=LioFNn7KrjQ

The magnet motor and the self-running magnet motor / generator systems from Dennis Danzik are presented on pages 193 - 239 in the German-language book "Autonome Magnetmotoren" about self-running magnet motors by Adolf and Inge Schneider , first edition June 2024 , A5 type format (15 cm x 21 cm) , 470 pages , publisher Jupiter-Verlag , ISBN 978--3-906571-47-8: https://www.jupiter-verlag.ch/shop/detail_neu.php?artikel=160&fromMain=j . See also https://gehtanders.de/produkt/autonome-magnetmotoren (in German).

The author of this website considers the fundamental operation principle of the self-running magnet motor / generator systems from Dennis Danzik in the same way as that of other magnet motors and related concepts which are presented further below on this home page. A special arrangement of magnetic fields, special magnetic effects, and in this case also a special triggering results macroscopically in an overall non-zero force and thus to a self-running motion. Microscopically it taps via magnetic fields and the spin of the electrons an everywhere present form of energy which is called space energy, vacuum energy, ether energy, or space-time energy. For further information about this energy form see this information document . See also papers which are presented further down on this home page.

Further references and information:

- Pages 3, 22, and 24 in the January / February 2023 issue of the NET-Journal (ISSN 1420-9292) (in German)

- German-language update on page 19 in the March / April 2021 issue of the NET-Journal (ISSN 1420-9292) (in German)

- Pages 25 - 53 in a German-language slide set (61 pages, 5 MB pdf) from a talk by Adolf Schneider about magnet motors on 3 October 2020 at a conference about novel energy technologies and related topics in Stuttgart in Germany. That slide set is also published in the website borderlands.de: www.borderlands.de/Links/Vortrag-Schneider-Magnetmotoren031020.pdf . See also page 22 in a German-language report in the November / December 2020 issue of the NET-Journal (ISSN 1420-9292) about the just mentioned conference (10 MB pdf, in German, see page 22): https://novam-research.com/resources/Kongress-Stuttgart_2-bis-4-Okt-2020_Bericht.pdf

- German-language report on pages 4 - 7 in the November / December 2019 issue of the NET-Journal (ISSN 1420-9292) (in German)

Entirely novel and environmentally friendly energy technologies

Power-magnifying electromagnetic AC generators without moving parts from the company Holcomb Energy Systems

Creation date 7 April 2023

Last update15 February 2026

Website of the company Holcomb Energy Systems: https://holcombenergysystems.com

Image source: Screenshot on 7 April 2023 from the website https://holcombenergysystems.com

Dr. Robert Holcomb and co-workers from the company Holcomb Energy Systems have developed power-magnifying electromagnetic AC generators without moving parts. They generate at least a doubled output power with respect to the input power. They are called in-line power generators (ILPGs) and the already existing systems generate an output power in the range from 3 kW to 500 kW. The specified input and / or output power of various Holcomb Energy Systems was confirmed by established certification companies such as SGS. The technology of the Holcomb Energy Systems is protected by several patents and patent applications in many countries. See e.g. WO2023118149A1 (4 MB pdf) which was published on 29 June 2023 and a paper from Adolf Schneider in German but with many citations in English. Before market launch it appears desirable to perform a field test and

study in more detail the dependence of the AC power amplification factor and other AC features like idle power, real power, and phase angle on various types of loads (ohmic, capacitive, and / or inductive). Also potential AC effects on the grid have to be considered.

Image source: Screenshot on 7 April 2023 from a video of Prof. Timothy Vaughan about novel energy technologies, namely https://www.youtube.com/watch?v=wvamIRUSbYE

During a German-language meeting of the Swiss Association for Space Energy (SVR) on 3 May 2024 in Zurich in Switzerland Adolf and Inge Schneider and others reported about a visit, measurements, and discussions in April 2024 at the company Holcomb Energy Systems in the United States. A German-language report about the visit, measurements, and discussions at the company Holcomb Energy Systems in the United States in April 2024 is published on pages 4 - 8 in the May / June 2024 issue of the German-language NET-Journal .

The Holcomb Energy Systems are presented on pages 253 - 280 in the German-language book "Autonome Magnetmotoren" about self-running magnet motors by Adolf and Inge Schneider , first edition June 2024 , A5 type format (15 cm x 21 cm) , 470 pages , publisher Jupiter-Verlag , ISBN 978--3-906571-47-8: https://www.jupiter-verlag.ch/shop/detail_neu.php?artikel=160&fromMain=j . See also https://gehtanders.de/produkt/autonome-magnetmotoren (in German).

On 14 October 2023 a congress about novel energy technologies took place in Zurich in Switzerland, see also the entry from 10 October 2023 under Events . Reinhard Wirth from the German-language blog https://gehtanders.de reported about his visit at the company Holcomb Energy Systems in the United States on 6 October 2023 [2-4], see also https://gehtanders.de/besuch-bei-holcomb-energy-systems . Ellen Sue Holcomb, Dr. Robert Holcomb, and an engineer from the company Holcomb Energy Systems joined via Zoom [2-4]. They presented their lab and their devices and some things about their research and development and reported about some ongoing and potential discussions and plans concerning further projects, series production, and marketing [2-4]. Their lecture was given in English and translated into German.

The research and development of the Holcomb Energy Systems started with rotating systems / generators / motors and it was attempted to weaken, circumvent or release the counter-electromotive force or back emf. The latter act like a braking force which limits the efficiency of the system. During their studies several things were discovered, developed and implemented, e.g.

- so-called Mu Metal was used to shield some magnetic fields
- the moving rotor was replaced by an arrangement of coils which create a rotating magnetic field by switching them sequentially. Initially the sequential switching of the coils was performed with solid state relays. However, they became often damaged by high currents. Later another way without using solid state relays was found and implemented.
- the coils windings display a special curved geometry and were so far manufactured manually
- the used electrical steel was shaped into a specific geometry. The so-called electrical steel is a ferromagnetic material which is magnetically soft, i.e. it can be easily magnetized and demagnetized
- the B versus H curve of the electrical steel in the coils is used only in a specific section and during various meetings two different operating modes were mentioned, namely one without zero-crossing (and that would be remarkable) and another one with zero-crossing. B is the so-called magnetic flux density of the ferromagnetic material and H is an applied external magnetic field which is usually generated by a coil.

The operating principle of the Holcomb Energy Systems is related to a special magnetic effect which is described in the patent DE3501076A1 from the former Siemens engineer Dr. Wolfgang Volkrodt, namely that the energy which is stored in a ferromagnetic material after its magnetization is significantly larger than the energy which was required to magnetize the ferromagnetic material. The patent DE3501076A1 is presented further down on this home page. Even if a partial connection between the Holcomb Energy Systems and Dr. Wolfgang Volkrodt's patent DE3501076A1 is more or less obvious, the outlined development of the Holcomb Energy Systems and their own patents and patent applications indicate that Dr. Robert Holcomb and his co-workers had an independent approach.

The author of this website considers the fundamental operation principle of the Holcomb Energy Systems in the same way as for other self-running or power-amplifying magnetic or electromagnetic systems. A special arrangement of magnetic

fields and special magnetic effects result macroscopically in an overall non-zero force and thus to a self-running motion or in an electric output power which is significantly larger than the electric input power. The latter allows also the construction of a self-running generator. Microscopically it taps via magnetic fields and the underlying spin of the electrons an everywhere present form of energy which is called space energy, vacuum energy, ether energy, or space-time energy. For further information about this energy form see this information document . See also papers which are presented further down on this home page.

The power-amplifying AC generators allow also the construction of a self-running or stand-alone generator system, namely by returning a part of the AC output power into the input of the AC power amplifier. The construction of a working self-running 15 kW generator system was already implemented in the lab of the company Holcomb Energy Systems. To decouple the AC input of the AC power amplifier from its AC output an AC / DC converter, a battery, and a DC / AC converter was used. However, the used type of battery became too hot and for a reliable and safe operation a proper energy storage subsystem is required.

References and further information:

[1] Report and presentation about a visit, measurements, and discussions in April 2024 at the company Holcomb Energy Systems in United States during a German-language meeting of the Swiss Association for Space Energy (SVR) on 3 May 2024 in Zurich in Switzerland. See also the entry from 15 April 2024 under Events (note that there are two entries from 15 April 2024)

[2] German-language reports and information in the German-language NET-Journal (ISSN 1420-9292):

- [2-6] Pages 18 - 20 in the January / February 2026 issue
- [2-5] Pages 4 - 8 in the May / June 2024 issue
- [2-4] Pages 16 - 18 in the November / December 2023 issue which are part of a German-language report about a German-language congress about novel energy technologies in Zurich in Switzerland on 14 October 2023
- [2-3] Pages 15 - 17 and 19 - 21 and also 11 - 15 in the September / October 2023 issue (in German)

- [2-2] Pages 16 - 18 in the July / August 2023 issue which are part of a German-language report about a German-language congress about novel energy technologies and related topics in Stuttgart in Germany on 17 and 18 June 2023 (in German): http://www.borderlands.de/net_pdf/NET0723S9-33.pdf . On 17 June 2023 Prof. Timothy Vaughan and Ellen Sue Holcomb & Dr. Robert Holcomb gave a presentation. Their lecture was given in English and translated into German. Prof. Timothy Vaughan was personally there and Ellen Sue & Dr. Robert Holcomb joined via Zoom [1-2]

- [2-1] Pages 4 - 7 in the May / June 2023 issue (in German): http://www.borderlands.de/net_pdf/NET0523S4-7.pdf

[3] The company Astra Energy Inc. promotes the Holcomb Energy Systems and there is an article in the Financial Times about it which was published on 4 October 2023: https://markets.ft.com/data/announce/detail?dockey=1330-8943368en-4SP7AIOUBDOC2JOC2UTLVRC3FL or download that article as pdf from this website: https://novam-research.com/resources/Astra-Energy-Inc-Announces-Strategic-Partnership-with-Sustainable-Energy-Technologies-Inc_Financial-Times_4-Oct-2023.pdf

[4] Zoom meeting with Ellen Sue Holcomb and Dr. Robert Holcomb during a meeting of the Swiss Association for Space Energy (SVR) on 18 August 2023 in Zurich in Switzerland

[5] Zoom meeting with Prof. Timothy Vaughan about the Holcomb Energy Systems during a meeting of the Swiss Association for Space Energy (SVR) on 28 April 2023 in Zurich in Switzerland

[6] German-language presentation from Adolf Schneider about the Holcomb Energy Systems: http://www.borderlands.de/Links/Holcomb-Energy-System.pdf

[7] About the Holcomb Energy Systems (HES) In-Line Power Generator (ILPG): http://www.borderlands.de/Links/Intro_to_ILPG.pdf

[8] German-language considerations by Adolf Schneider about the energy / power amplification of the Holcomb Energy Systems: http://www.borderlands.de/Links/Holcomb-Energie.pdf

[9] Interview with Robert Holcomb from April 2022 - Green Tech: Dr Robert Holcomb of Holcomb Energy Systems On How Their Technology Will Make An Important Positive Impact On The Environment (in English and translated into German): http://www.borderlands.de/Links/Interview%20mit%20Green%20Tec.pdf

[10] List with links to many videos about the Holcomb Energy Systems: http://www.borderlands.de/Links/Holcomb-Filme.pdf

[11] Video from Prof. Timothy Vaughan about entirely novel energy technologies which are based on special magnetic effects and the electron spin: https://www.youtube.com/watch?v=wvamIRUSbYE . Keywords: Inventions, conventional and unconventional energy technologies, various inventors / researchers / scientists and their unconventional generators, Nikola Tesla, thermodynamics, magnetism, special magnetic effects, electron spin, Holcomb Energy Systems

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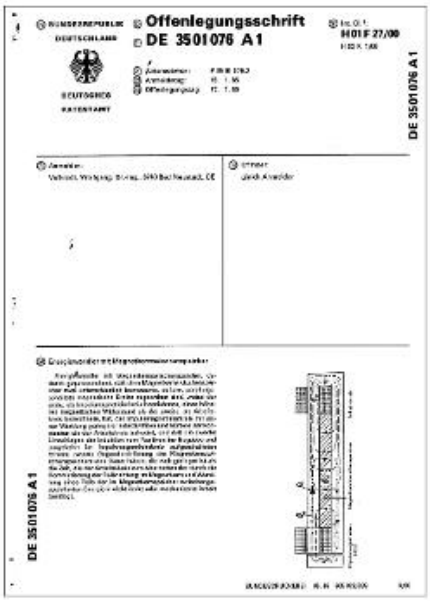
Entirely novel and environmentally friendly energy technologies

Patent DE3501076A1 "Energy converter having a magnetic-core intermediate store" from 1986 by the former Siemens engineer Dr. Wolfgang Volkrodt

Creation date 8 August 2024

For the English version of this patent see for example https://patents.google.com/patent/DE3501076A1/en . The text is a translation from the original German-language patent specification.

The original German-language patent specification including figures is for example available via this website: https://novam-research.com/resources/Patent-DE3501076A1.pdf . According to the description the energy which is stored in a ferromagnetic material after its magnetization is significantly larger than the energy which is necessary to



Page 1 (top) and Figure 2 (bottom) from the original

magnetize the ferromagnetic material. This phenomenon can be used to construct an electromagnetic device which provides at its output an amount of usable electric energy or power which is significantly larger than the energy or power which is fed into its input.

German-language patent DE3501076A1

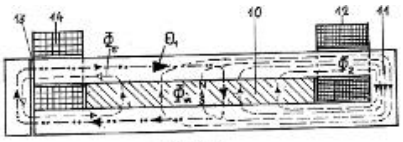


Fig. 2

A German-language report about that congress is published on pages 4 - 21 in the November / December 2025 issue of the German-language journal NET-Journal (ISSN 1420-9292). It comprises on pages 18 and 19 the German-language sub-report "Revolutionärer hydraulisch-elektrischer Generator bis 1 MW" about the generators developed by Dr. V. Marukhin and his co-workers.

The following table from page 44 of the slide set from Adolf Schneider displays an overview of the self-running generators which are based on autonomous undamped hydraulic oscillations which were developed by Dr V. Marukhin and his co-workers (as of August 2025) (there and in the slide set from Adolf Schneider the generators are called HEG which stands for Hydraulic Energy Generator and "HEG-Typ" means HEG type):

Citations from the English patent specification:

" ... Let us briefly summarize the current state of knowledge: The necessary energy to magnetize or reverse magnetize a magnetic core store is considerably smaller than the energy content that can be measured afterwards in the permanent magnet ... Those to be expected according to previous theories of energy technology Alternating hysteresis losses are a multiple of what is actually required for reversal of magnetization supplied electrical energy. The theory does not agree with either the law of conservation of energy still match experimental experience ...

... Up to now, conventional physics has only known the real components of electricity, magnetism and gravity. In order to interactions between the aforementioned quantities To be able to describe it in terms of vibrations and eddies, however, is more complex Sizes. According to more recent theories, imaginary quantities must also be in our space as are assumed to be present, from which, under certain conditions, a transformation is possible in forms of energy known to us ...

... According to the above hyperenergy theory, one understands by ferromagnetism the alignment of unpaired electrons (they are small hyperspace vortices) around it to establish the resulting magnetic vector field ...

... Apparatus and machines that use hyperspace as an energy source use are not perpetual motion machines, but rather energy converters in the conventional sense ..."

Entirely novel and environmentally friendly energy technologies

Power-amplifying and self-running electromagnetic generators without moving parts from the company SEMP Research Institute Co. Ltd.

Creation date 7 July 2024

Last update 7 March 2026

The company SEMP Research Institute Co. Ltd. has developed power-amplifying and self-running electromagnetic generators without moving parts. They are called AISEGs which stands for Artificial Intelligence Smart Electromagnetic Generator.

Website of the company SEMP Research Institute Co. Ltd.: https://www.semp.or.kr/en

AISEG at the Expo COP28 in Dubai in December 2023. On the right there are overall 60 lamps which can be used as load at the output of the AISEG. Image source: Screenshot on 7 July 2024 from https://www.semp.or.kr/en

Short videos (a few minutes) about the AISEG: https://www.youtube.com/watch?v=SLJH8Ad2b44 from December 2023 and https://www.youtube.com/watch?

v=VgsPcGvLyn4 from
December 2018

The input of an AISEG is fed with a pulsed DC current and its output consists of an AC current. The AC output power is up to 17 times higher than the pulsed DC input power. A self-running operation becomes possible when a part of the AISEG output energy or power is fed into its input. It is elaborate to find the proper adjustments which allow a self-running operation. So far a self-running operation was tested up to 6 hours.

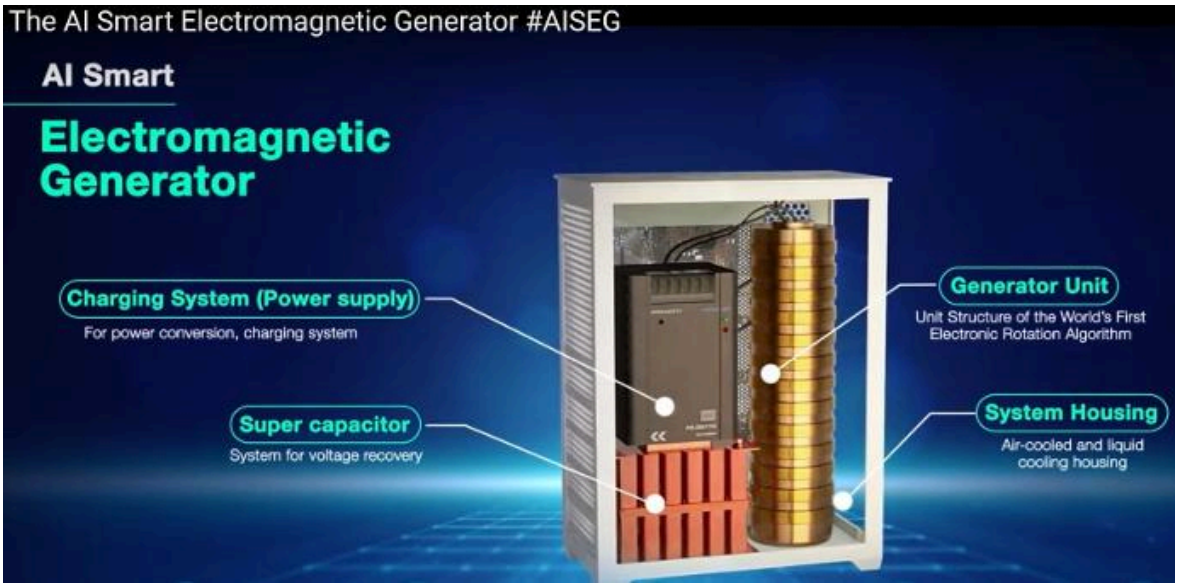


Image source: Screenshot on 8 July 2024 from video https://www.youtube.com/watch?v=SLJH8Ad2b44 . For a better readability the text of the 4 labels is repeated here: Charging System (Power supply) - For power conversion, charging system | Supercapacitor - System for voltage recovery | Generator Unit - Unit Structure of the World's First Electronic Rotation Algorithm | System Housing - Air cooled and liquid cooling housing. Note: So far a part of the AISEG output energy is fed via an AC / DC converter and a battery into its input. In future AISEG constructions it is intended to use the sketched supercapacitor instead of a battery.

On the right: Woo Hee Choi, co-founder and CEO of SEMP and main co-inventor of the AISEG, on 23 June 2024 at a German-language conference about novel energy technologies and related topics nearby Stuttgart in Germany. Overall seven staff members from SEMP were there and presented a talk about the AISEG. There was a translation from Korean into English and from English into German. A German-

language report about (this conference and) the AISEG is published on pages (4) 21- 23 (25) in the July / August 2024 issue of the German-language NET Journal (ISSN 1420-9292).

Image source: Newsletter from June 2024 from the Austrian Association for Space Energy (ÖVR)

In August 2024 eleven people from Switzerland, Germany, and other countries have visited SEMP and their staff members in South Korea and measurements were performed at an AISEG. In contrast to the latest AISEG which was presented in Dubai in December 2023 the available AISEG at SEMP in August 2024 was one of the previous models. At the demonstration of the available AISEG the battery became very hot and a reliable measurement of the input power was not possible. There are two German-language reports about the visit at SEMP in August 2024, namely

- on pages 4 - 14 in the September / October 2024 issue of the German-language NET Journal (ISSN 1420-9292) (in German): http://www.borderlands.de/net_pdf/NET0924S4-14.pdf
- in the blog gehtanders.de: https://gehtanders.de/semp-generator-die-reise-nach-korea (in German but that webpage is equipped with a Google translator)

In October 2024 four people, three from Switzerland and one from Germany, have visited SEMP and their staff members in South Korea. The results of this second visit at SEMP

- were presented at a German-language convention about novel energy technologies on 30 November 2024 in Zurich in Switzerland, see entry from 29 November 2024 under events and pages 10, 11, and 14 in a German-language report about that convention in the January / February 2025 issue of the German-language NET Journal (ISSN 1420-9292) (in German)
- are published in a German-language report on pages 6 - 9 in the November / December 2024 issue of the German-language NET Journal (ISSN 1420-9292) (in German). In the same issue there are three further German-language reports about SEMP and the attempts to promote the AISEG, namely on pages 10 - 19

As of December 2024: The latest AISEG is still located and locked in Dubai because SEMP was after its demonstration at the Expo COP28 in December 2023 in Dubai not able to pay the stand fee. About 2 million USD are required to unleash that AISEG

from Dubai and transport it back to SEMP in South Korea. SEMP and their staff members need further funding to develop / provide an AISEG prototype which allows a verifiable self-running operation with various types of loads (ohmic, capacitive, and / or inductive) over an extended period of time. A part of the AISEG output energy is so far fed via an AC / DC converter and a battery into the input of the AISEG. This feedback system, which decouples the input from the output, needs particular attention. It appears desirable to

- decrease the amount of energy which is branched off from the output and fed into the input
- replace the battery by supercapacitors or another appropriate system

As of December 2025 (from a German-language update which is published on page 23 in the January / February 2026 issue in the German-language NET Journal / ISSN 1420-9292): An engineer team from United States has performed a comprehensive inspection of an AISEG including an 11 hours long endurance test. The results indicate that the AISEG technology is suitable for commercialization and an initial public offering (IPO). Further inspections and endurance tests by another engineers are planned, among others a 48 hours long load test. Furthermore, that AISEG which was locked in Dubai is meanwhile available again and can be used again for demonstration and testing purposes.

The technology of the AISEG is scalable, i.e. generators with an output power from 5 kW to several 100 kW and more can be manufactured. The gravimetric power density of an AISEG is currently about 28 W / kg.

The operating principle of the AISEG is related to the AC power amplifiers from the company Holcomb Energy Systems and the patent DE3501076A1 from the former Siemens engineer Dr. Wolfgang Volkrodt. The latter is based on the observation hat the energy which is stored in a ferromagnetic material after its magnetization is significantly larger than the energy which was required to magnetize the ferromagnetic material. The AC power amplifiers from the company Holcomb Energy Systems and the patent DE3501076A1 from Dr. Wolfgang Volkrodt are presented further up on this home page. Even if a partial connection between the AISEG and the Holcomb Energy Systems as well as Dr. Wolfgang Volkrodt's patent DE3501076A1 is more or less obvious, the AISEG design, patents, and patent applications indicate that the AISEG was developed in an independent manner. The AISEG comprises several distinct features such as the use of

- several coil towers whereby each tower consists of a vertical arrangement of alternating input and output coils which are magnetically connected by a common ferromagnetic core which is magnetically soft (this arrangement corresponds to an open transformer)
- a pulsed DC current which is fed into the input coils to magnetize and demagnetize the ferromagnetic core material which is magnetically soft
- the so-called bandwagon effect which means that several adjacent coil towers couple and amplify each other their impact via their magnetic stray field
- artificial intelligence (AI) to create the control software of the generator

For the AISEG technology more than 70 patents were applied in 60 countries. A patent application which describes the fundamentals is for example the United States patent application publication US20230198368A1 which was published on 22 June 2023 and can be downloaded from this website: https://novam-research.com/resources/United-States-Patent-Application-Publication-US20230198368A1_2023.pdf

The author of this website considers the fundamental operation principle of an AISEG in the same way as for other self-running or power-amplifying magnetic or electromagnetic systems. A special arrangement of magnetic fields and special magnetic effects result macroscopically in an overall non-zero force and thus to a self-running motion or in an electric output power which is significantly larger than the electric input power. The latter allows also the construction of a self-running generator. Microscopically it taps via magnetic fields and the underlying spin of the electrons an everywhere present form of energy which is called space energy, vacuum energy, ether energy, or space-time energy. For further information about this energy form see this information document . See also papers which are presented further down on this home page.

The SEMP Research Institute Co. Ltd. and their AISEG is presented on pages 240 - 252 in the German-language book "Autonome Magnetmotoren" about self-running magnet motors by Adolf and Inge Schneider , first edition June 2024 , publisher Jupiter-Verlag , ISBN 978--3-906571-47-8: https://www.jupiter-verlag.ch/shop/detail_neu.php?artikel=160&fromMain=j . See also https://gehtanders.de/produkt/autonome-magnetmotoren (in German).

Further references and information:

A report about the AISEG was presented during a German-language meeting of the Swiss Association for Space Energy (SVR) on 19 July 2024 in Zurich in Switzerland. German-language program of this meeting: https://svrswiss.org/pdfs_aso/Programm73.SVR-Meeting190724.pdf

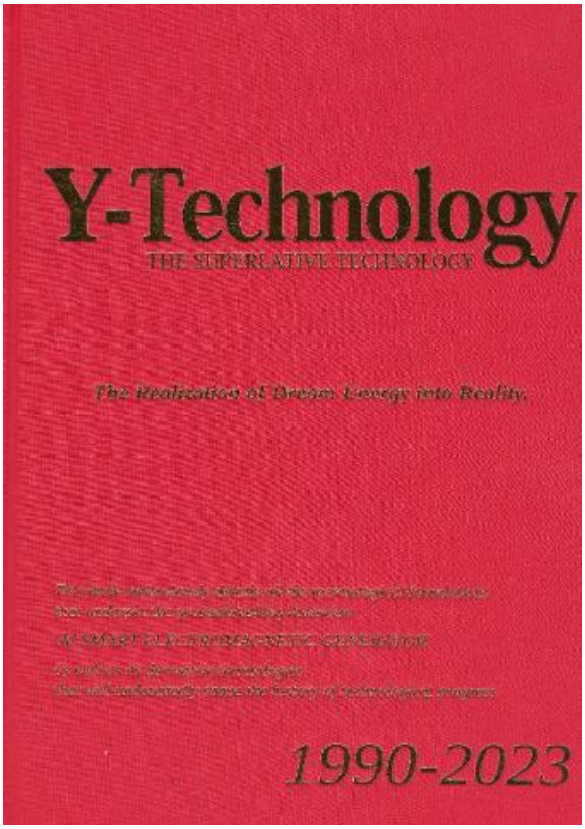
A German-language report about the AISEG is published on pages 9 - 12 in the May / June 2024 issue in the German-language NET Journal (ISSN 1420-9292) (in German).

Brochure about the AISEG

Paper size A4

Publisher: SEMP Research Institute Co. Ltd.

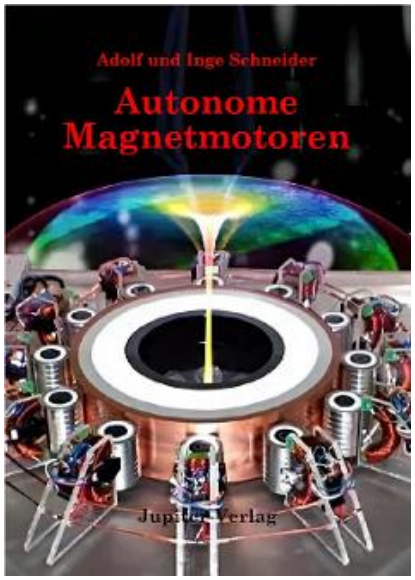
Book "Y-Technology". Paper size A4 and 1532 pages. Weight about 5 kg. It describes several inventions, especially the AISEG. It has no ISBN number and the publisher is the company SEMP Research Institute Co. Ltd. It can be requested and purchased via SEMP and its price is 1000 USD.



Entirely novel and environmentally friendly energy technologies

Various types of self-running magnet motors

Last update 5 October 2025



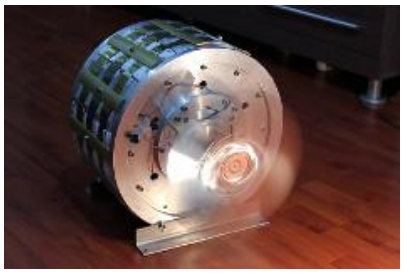
German-language book "Autonome Magnetmotoren" about self-running magnet motors by Adolf and Inge Schneider , first edition June 2024 , A5 type format (15 cm x 21 cm) , 470 pages , publisher Jupiter-Verlag , ISBN 978--3-906571-47-8: https://www.jupiter-verlag.ch/shop/detail_neu.php?artikel=160&fromMain=j . See also https://gehtanders.de/produkt/autonome-magnetmotoren (in German). Presented in this German-language book are also the self-running magnet motor / generator systems from the company Quantum Energy (Inductance Energy) Corporation,

the solid state power amplifiers from the company Holcomb Energy Systems, the self-running solid state electromagnetic generator from the company SEMP Research Institute Co. Ltd., and the kinetic power plants from the company Rosch Innovations / Save The Planet AG which comprise specially modified electromagnetic generators. These systems are also presented on this home page and in this website.

German-language slide set of a talk by Adolf Schneider about magnet motors on 3 October 2020 at a conference about novel energy technologies and related topics in Stuttgart in Germany (61 slides or pages, 5 MB pdf). This slide set is also published in the website borderlands.de: www.borderlands.de/Links/Vortrag-Schneider-Magnetmotoren031020.pdf

Self-running magnet motor from the Turkish inventor Muammer Yildiz

Website: https://hmsbturk.com



Previous image from the website https://hmsbturk.com

A presentation event with a self-running permanent magnet motor took place on 17 and 18 December 2022 in Balcova at Izmir in Turkey. German-language reports about this event and related topics are published on pages 4 - 8 in the January / February 2023 issue and on pages 7 - 9 in the March / April 2023 issue of the NET-Journal (ISSN 1420-9292) (in German). A report, pictures, and videos about this event and related topics was presented during a German-language meeting of the Swiss Association for Space Energy (SVR) about novel energy technologies on 10 February 2023 in Zurich in Switzerland (in German): https://svrswiss.org/pdfs_aso/Programm64.SVR-Meeting100222.pdf .

A self-running prototype of the Yildiz Motor was presented and disassembled at the Eindhoven University of Technology in the Netherlands in April 2009 and at the University of Delft in the Netherlands in May 2010: https://www.youtube.com/watch?v=mHW6b1aFPfU

in July 2019 the first part of an intended series of live-streamed tests of a Yildiz Motor was performed [11]. The setup at the company D'Ambros in Belluno in Italy comprised among various equipment a 7,5 kW Yildiz Motor which propels a generator. At first the self-running and zero emission Yildiz Motor operated well for about 7 hours and this can be already considered a success. After about 7 hours a technical defect did arise and the Yildiz Motor had to be stopped.

Several attempts to advance the Yildiz Motor towards commercial applications were not successful because discussions between Muammer Yildiz and potential investors did not result in mutual agreements.

The author of this website considers the fundamental operation principle of the Yildiz Magnet Motor and other self-running magnet motors in the following way: A special arrangement of magnetic fields results macroscopically in an overall non-zero force and thus to a self-running motion. Microscopically it taps via magnetic fields and the underlying spin of the electrons an everywhere present form of energy which is called space energy, vacuum energy, ether energy, or space-time energy. This energy form is presented in an information document about entirely novel and environmentally friendly energy technologies. See also some of the papers which are presented on this home page further down.

The Yildiz Motor comprises several hundred permanent magnets. The self-running motion comes about only if all permanent magnets are firmly and very precisely fixed at specific positions. That seems to be not easy and quite challenging. During operation the mechanical vibrations may change the position of one or several permanent magnets if they are not very firmly positioned. Furthermore, some permanent magnets are connected with a spring and their positions are periodically moved in a jumpy way. Such periodic mechanical concussions of the permanent magnets deteriorate stepwise their magnetic performance.

Other self-running magnet motors and related concepts

There are other magnet motor concepts which achieve a self-running motion with a significant less number of permanent magnets [1,2,3,4,5]. About such a concept was reported during a meeting of the Swiss Association for Space Energy (SVR) on 14 October 2022 in Zurich in Switzerland. A team of researchers and engineers have performed very extensive and detailed computational simulations by using the laws of conventional electrodynamics and the so-called finite element method. The results indicate a self-running motion with a system that comprises only 14 permanent magnets. A motor with a diameter of about 40 cm and a length of about 60 cm should generate a mechanical output power of about 20 kW. The intended manufacturing of such a system will reveal if the obtained computational results are correct. So far this project is not (yet) published but somewhat about it is presented in Ref. [1].

There is also a self-running magnet motor for which remarkably a patent was granted, namely the US patent US8487484B1 . The specified inventors are Harold L. Miller, Jr. and Andrew E. Colson, Jr. The correponding patent application was filed in 2012 and the granted application was published in 2013. Its current status (October 2025) is displayed as expired and the specified reason is fee-related. This type of magnet motor is presented in the following video which was published in January 2025: https://www.youtube.com/watch?v=JF-OZZC9oQ8 . The video title is "The World's 1st Magnet Motor". That is certainly not true because, for example, the German mechanical engineer Friedrich Lüling has already around 1954 invented a magnet motor. See, for example, the German-language video https://www.youtube.com/watch?v=r0mP4qGDwols

Another example of a magnet motor concept and open source project is the so-called Kryon Engine: https://www.kryonengine.org

References, papers, presentations, and reports

[1] German-language report by Adolf Schneider about various magnet motor concepts on pages 35 - 38 in the November / December 2022 issue of the NET-Journal (ISSN 1420-9292) (in German)

[2] Paper "Essay on Magnetic-Wind Mills - Part 1: Analysis and Design" from 2019 by Dr. J. L. Duarte from the Department of Electrical Engineeering of the Eindhoven University of Technology in the Netherlands: https://doi.org/10.6100/94FC4756-AC0A-4A9C-A1DD-58530E694BB8 . A version of Part 2, namely that from 15 November 2022, can be downloaded from this website: https://novam-

research.com/resources/Essay-on-Magnetic-Wind-Mills_Part2_J-L-Duarte_v4_15-Nov-2022.pdf

[3] Paper "Design and Simulation of Self-Running Magnetic Motor" from 2017 by Abdul Halim Ali and Ahmad Najmuddin Che Ismail: https://www.researchgate.net/publication/335257538 or download it as pdf from this website: https://novam-research.com/resources/Design-and-Simulation-of-Self-Running-Magnetic-Motor_A-H-Ali-and-A-N-C-Ismail_2017.pdf

[4] Paper "Design and Simulation of Free Energy Permanent Magnet Motor (FEPMM)" from 2016 by Amel Ridha: https://www.researchgate.net/publication/307513635

[5] The following paper from 2015 presents in its introduction various types of magnet motors and reports about an own concept: Design of open source straight permanent magnet motor, Amel Ridha and Haider H. Jabber, Kufa Journal of Engineering, Volume 6 (2015), pages 24 - 38: https://doi.org/10.30572/2018/KJE/621146 . Also published via https://www.researchgate.net/publication/307513641

[6] Two papers "Introducing the Yildiz Motor" and "Modeling the Yildiz Motor" from 2010 and 2012 by Dr. J. L. Duarte from the Department of Electrical Engineeering of the Eindhoven University of Technology in the Netherlands: https://novam-research.com/resources/Introducing-and-modeling-the-Yildiz-Motor_v01.pdf

[7] A review on how a Perpetual Motion Machine generates electrical power, M. N. Hidayat, S. P. Chairandy, and F. Ronilaya, IOP Conference Series: Materials Science and Engineering 1098 (2021) 042063 (1 - 7) https://doi.org/10.1088/1757-899X/1098/4/042063

[8] Updates about magnet motors are presented on page 14 in a German-language report in the November / December 2020 issue of the NET-Journal (ISSN 1420-9292) about a conference from 2 to 4 October 2020 in Stuttgart in Germany about novel energy technologies and related topics (10 MB pdf, in German, see page 14): https://novam-research.com/resources/Kongress-Stuttgart_2-bis-4-Okt-2020_Bericht.pdf

[9] German-language slide set (61 pages, 5 MB pdf) of a talk by Adolf Schneider about magnet motors on 3 October 2020 at a conference about novel energy

technologies and related topics in Stuttgart in Germany . This slide set is also published in the website borderlands.de: www.borderlands.de/Links/Vortrag-Schneider-Magnetmotoren031020.pdf

[10] German-language report about the Yildiz Motor on page 20 in the September / October 2020 issue of the NET-Journal (ISSN 1420-9292) (in German)

[11] A German-language report about a test of a Yildiz Motor is published on pages 4 - 7 in the July / August 2019 issue of the NET-Journal (ISSN 1420-9292) (in German): http://www.borderlands.de/net_pdf/NET0719S4-7.pdf

[12] The Yildiz Motor is presented on pages 16 - 19 in a slide set (35 pages, 2 MB pdf) of a talk by Adolf Schneider about novel energy technologies on 3 November 2016 at the World Clean Energy Conference 2016 in Geneva in Switzerland

[13] The Yildiz Motor was presented at the 41th inventor fair in Geneva in Switzerland, see page 12 in the following German-language report in the May / June 2013 issue of the NET-Journal (in German): http://www.borderlands.de/net_pdf/NET0513S4-14.pdf

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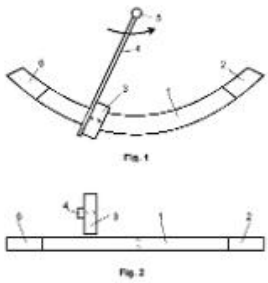
Entirely novel and environmentally friendly energy technologies

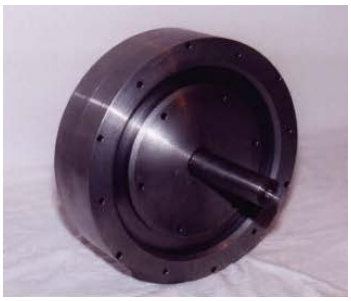
Self-running cryogenic magnet motor from Walter Thurner

Last update 15 August 2021

Please note: Walter Thurner died on 12 February 2021 and meanwhile his former German-language website www.walter-thurner.de no longer exists and his former German-language pdf document www.walter-thurner.de/magnet.pdf is no longer available via his former website. However, his original pdf document is available via this website: https://novam-research.com/resources/Walter-Thurner_Kryo-Magnet-Motor.pdf . The author of this website thanks Walter Thurner for his remarkable work and his former pdf document whose essential parts are presented in English in this report. A German-language obituary is published on pages 63 and 64 in the May / June 2021 issue of the German-language NET-Journal (ISSN 1420-9292).



Images from former website www.walter-thurner.de



Report about the self-running cryogenic magnet motor of Walter Thurner

This magnet motor represents a nice example of a self-running system which can generate usable energy. Its operation principle appears fairly comprehensible. Once it is understood one has an idea how a self-running system can work concretely, at least at the macroscopic scale. It appears worthwhile and instructive to try to recreate this magnet motor and develop a system that limits and controls its speed of rotation.

Entirely novel and environmentally friendly energy technologies

Reactionless electromagnetic generator from Paramahamsa Tewari

Last update 22 February 2022

Report about Paramahamsa Tewari, his reactionless electromagnetic generator, and his Space Vortex Theory

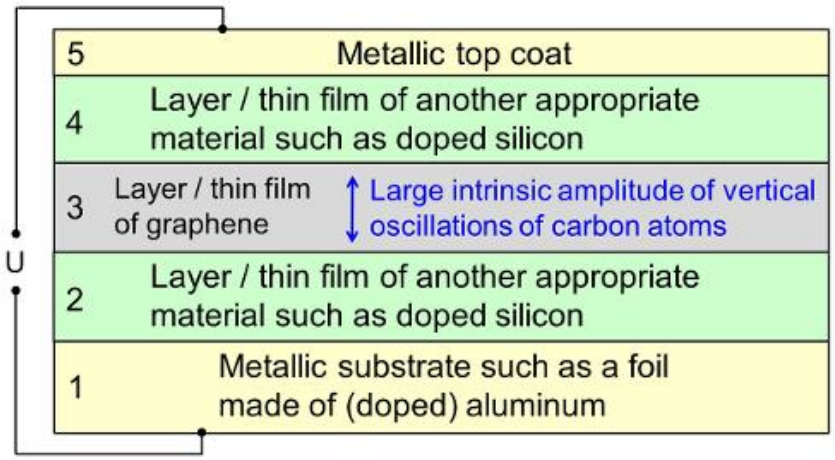
Image from https://www.tewari.org

Entirely novel and environmentally friendly energy technologies

So-called neutrino voltaic / cells / technology - Self-running and zero emission electrical DC generators without moving parts from the company Neutrino

Last update 7 June 2025

Report about the so-called neutrino voltaic



Entirely novel and environmentally friendly energy technologies

Reid Cell or Crystal Cell

Last update 5 June 2025

The Reid Cell or Crystal Cell provides a small amount of electrical DC power of about 1 mW over a very long period of time, so far since 1998. This and some other features cannot be explained solely by conventional electrochemical effects. Video from 2023 about the Reid Cell or Crystal Cell by Eckhard Kantz and Marcus Reid: https://www.dailymotion.com/video/x93j7o4 . Websites: https://vakuumenergie.de and https://quantumpowermunich.de . For comments on the Reid Cell or Crystal Cell see https://quantumpowermunich.de/wp-content/uploads/2023/12/Comments-Reid-Cell.pdf and / or https://vakuumenergie.de/wp-content/uploads/2023/12/Comments-Reid-Cell.pdf

Image source: See for expample https://vakuumenergie.de

The Reid Cell or Crystal Cell represents an interesting topic for further research, development, and advancement, especially with respect to the origin of the electrical power and how the power can be increased.

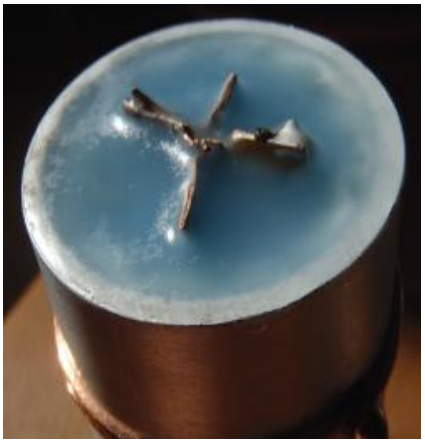


Image source: See for example Screenshot from former video https://www.dailymotion.com/video/x8lkslc from March 2024

A book about the Reid Cell or Crystal Cell is already available or coming soon:

- Title: Reid Cell
- Authors: Marcus Reid, Eckhard Kantz, and Dr. Nigel Deyer
- https://vakuumenergie.de
- https://vakuumenergie.de/book
- https://www.kulturkaufhaus.de/en/detail/ISBN-9783982213729/Reid-Marcus/Reid-Cell

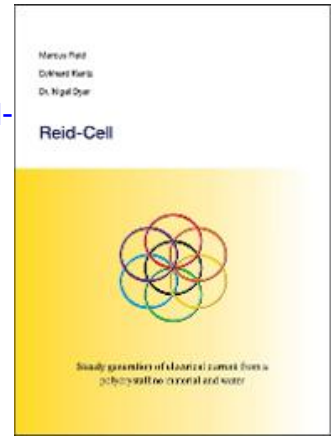


Image source: See for expample https://vakuumenergie.de

For further information like front cover, back cover, foreword, and contents see https://quantumpowermunich.de/wp-content/uploads/2024/04/Marcus-CC-Innenseiten-Eriks-Edits.pdf and / or https://vakuumenergie.de/wp-content/uploads/2025/03/Marcus-CC-Innenseiten-book-prev.pdf

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Entirely novel and environmentally friendly energy technologies

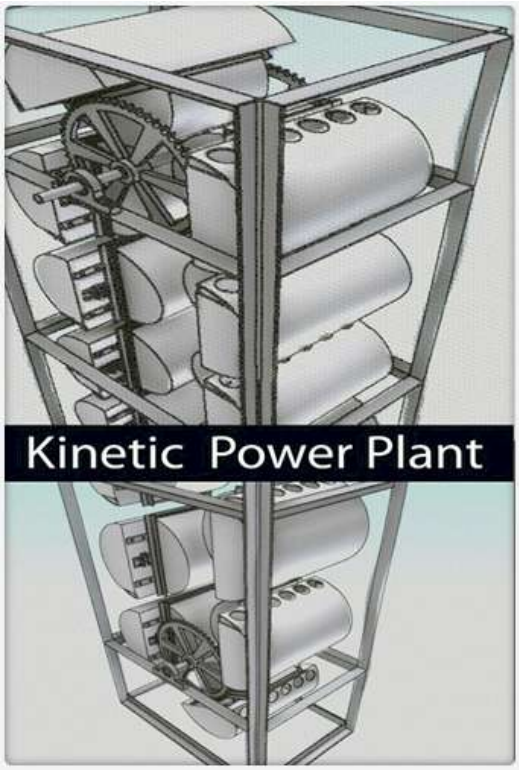
Self-running and zero emission power plants from the company Rosch Innovations

Last update 7 October 2025

Updates to the Report and the Abstract / Summary which is presented further down:

- In 2025 the main owner of the company Rosch Innovations died. Presently it seems uncertain how the company will proceed and if their power plants continue to be available

- On 19 September 2025 Robert Schrade presented at a meeting of the Swiss Association for Space Energy (SVR) in Zurich in Switzerland a German-language talk about his buoyancy or kinetic power plant. He presented himself as the inventor of the buoyancy or kinetic power plants of the company Rosch Innovations. The name of Robert Schrade's former and present company is eurosch GmbH and it is located in Germany. There is a German patent application DE102009043356A1 from Robert Schrade from 2009. It was published in 2011 and is meanwhile withdrawn. Robert Schrade reported that he has around 2009 developed and constructed two prototypes which were not self-running because they were not equipped with a generator. Around that time a disagreement between him and his colleague(s) or co-worker(s) did arise. The other one or some of the others went then a different way. Robert Schrade communicated that a self-running system requires a special design and way how to inject the air and a special control of the overall system. And it works with an appropriate conventional generator. He is convinced that he



Sketch of the buoyancy part of a kinetic power plant (KPP). It comprises a water column which contains a chain drive, fillable containers which are attached at the chain drive, and a valve system which can fill the containers with air. Image from the Rosch Innovations website https://rosch.ag . Not shown in this picture are other parts of the KPP like an electric generator, an electrically-operated compressor, and a control unit.

is able to build a self-running prototype. For a self-running prototype which generates an electrical output power of some kW the estimated amount is about 30000 Euro plus the costs for the working hours

Abstract / Summary of the report: The company The company Rosch Innovations / Save The Planet AG has developed so-called kinetic power plants (KPPs). These self-running and zero emission power plants generate permanently usable electric energy without any fuels, wind power or solar energy. Rosch offers via distribution partners industrial size power plants with an electric output power of 200 kW and 5, 40, and 100 MW as well as 1 MW and from 5 MW with fully scalable 500 kW modules . Distribution partners are e.g. the global engineering, procurment, and construction company Ki-TECH for the KPP technology and the German company E-Cat Deutschland GmbH . An industrial size power plant at a publicly known and visitable customer site does not yet exist. However, potential industrial customers or potential licensees can visit and inspect an operating 100 kW and 500 kW power plant, provided that they provide a letter of intent and a proof of funds. The 100 kW and 500 kW power plants are located at a Rosch site in Thailand where they power that site and feed electricity into the grid, see the following video from 2022: https://www.youtube.com/watch?v=3IdNow8vZdg . An inspection by an engineer team of the potential customer or potential licensee allows a verification of the genuineness of the power plants

Picture from the 100 kW demo plant in Thailand. Image copied via screenshot on 6 June 2022 from the Ki-TECH website https://www.ki-tech.global/home-login which comprises also other pictures

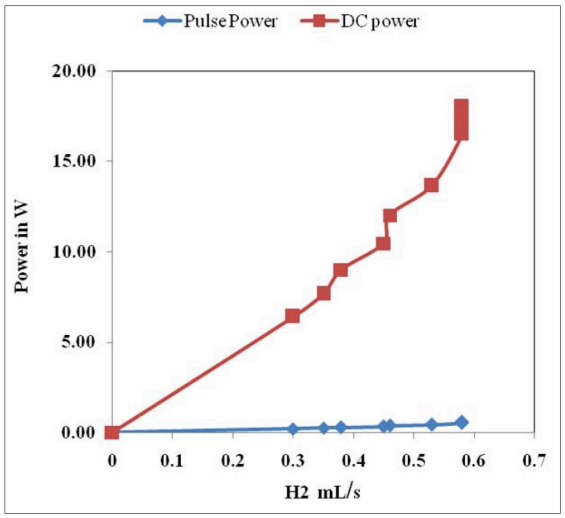
and rule out possible objections such as a hidden power source. See, for example, the results of a technical examination of the 100 kW power plant in Thailand from 2018: http://borderlands.de/Links/KPP-Messung061218-Wunder+Holper.pdf . As already mentioned, an industrial size power plant at a publicly known and visitable customer site does not yet exist. However, globally some power plants are under construction, see Ref. [7r] in the report, and there are some large projects which are in the planning stage, see e.g. the website https://wvge-me.com which presents projects and a roadmap for Montenegro and other Balkan states. Pictures on the right: Sketch of the buoyancy part of a kinetic power plant, the 100 kW demo plant in Thailand, and a former 60 kW pilot plant. An often asked question is from where the generated energy comes from. The author of this website assumes that this novel energy technology is just one type of many possible types of systems which extract usable energy from an everywhere present form of energy which is called space energy, vacuum energy, zero-point energy, or ether energy as presented in this information document . Papers about such an advanced and little-known approach are also published in established science journals, see e.g. Refs. [11-14] in the report or the papers which are presented further down on this home page.

Picture from a 60 kW pilot plant from a publication in the NET Journal (ISSN 1420-9292), issue May / June 2016, page 9 (in German)

Entirely novel and environmentally friendly energy technologies

Water splitting by pulsed electrolysis

Last update 12 May 2023



The figure displays the effect of a conventional DC electrolysis and a pulsed electrolysis on the hydrogen production rate.
Image source: Figure 6 in the paper C. H. Dharmaraj and S. AdishKumai, International Journal of Energy and Environment 3 (2012) 129 - 136

In established physics / chemistry / science it is assumed that the modulus of the energy which is required to split a water molecule (electrolytically) into hydrogen and oxygen is always equal to the modulus of the energy which is released when hydrogen and oxygen recombine into a water molecule. This is because the two hydrogen atoms and the one oxygen atom in a water molecule are bound together by electric or electromagnetic forces which are considered as so-called conservative force fields, i.e. a closed loop motion of a particle in such a force field can never result in a gain of usable energy.

However, many researchers and inventors have claimed that under specific conditions it is possible to split water into hydrogen and oxygen with much less energy than usual. In this case usable energy can be generated when the created hydrogen and oxygen recombine into water with a release of the usual amount of energy.

The following work is a published example how water can be split into hydrogen and oxygen with much less energy than usual, namely by a special pulsed electrolysis:

Economical hydrogen production by electrolysis using nano pulsed DC
C. H. Dharmaraj and S. AdishKumar , International Journal of Energy and Environment , Volume 3 (Issue 1), 2012, pages 129 - 136 , https://www.ijee.ieefoundation.org/vol3/issue1/IJEE_13_v3n1.pdf . or open or download this paper as pdf from this website.

The authors of this paper have not discussed the potential physics behind this remarkable phenomenon. The author of this website assumes that any physical process can be performed / modified / designed in such a way that it taps an everywhere present energy source which is called space energy, vacuum energy, ether energy, or space-time energy. For further information about this form of energy see this information document about entirely novel and environmentally friendly energy technologies. From this perspective it is assumed that a specifically pulsed electrolysis results somehow in an assisted water splitting where a large part of the required splitting energy is provided by an usable inflow of space / vacuum / ether / space-time energy.

Further related information: Adolf Schneider has prepared a German-language overview and compilation about various conventional and unconventional water splitting processes and their efficiency (in German): http://www.borderlands.de/Links/Wasserstoffherstellung.pdf

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Advanced physics and entirely novel and environmentally friendly energy technologies - International Space Federation (ISF)

Last update 27 October 2025

Image from a screenshot of https://www.spacefed.com on 21 January 2024

Website: https://www.spacefed.com

YouTube channel: https://www.youtube.com/@intlspacefed

Swiss corporation with headquarters in Geneva and laboratories in France
Research and development in the field of quantum vacuum, gravity and unified physics which also includes entirely novel and environmentally friendly energy technologies. Research director is Nassim Haramein

Significant paper "The Origin of Mass and the Nature of Gravity" by Nassim Haramein, Cyprien Guermonprez, and Olivier Alirol - published on 26 September 2023 via zenodo.org: https://doi.org/10.5281/zenodo.10125315 . Abstract of the paper and video about the paper:

- Abstract of the paper: From the early explorations of thermodynamics and characterization of black body radiation, Max Planck predicted the existence of a non-zero expectation value for the electromagnetic quantum vacuum energy density or zero-point energy (ZPE). From the mechanics of a quantum oscillator, Planck derived the black body spectrum, which satisfied the Stefan-Boltzmann law with a non-vanishing term remaining where the summation of all modes of oscillations diverged to infinity in each point of the field. In modern derivation, correlation functions are utilized to derive the coherent behavior of the creation and annihilation operators. Although a common approach is to normalize the Hamiltonian so that all ground state modes cancel out, setting artificially ZPE to zero, zero-point energy is essential for the mathematical consistency of quantum mechanics as it maintains the non-commutativity of the creation and annihilation operators resulting in the Heisenberg uncertainty principle. From our computation, we demonstrate that coherent modes of the correlation functions at the characteristic time of the proton correctly result in the emergence of its mass directly from quantum vacuum fluctuation modes. We find as well that this energy value is consistent with a Casimir cavity of the same characteristic distance. As a result, we developed an analytical solution describing both the structure of quantum spacetime as vacuum fluctuations and extrapolate this structure to the surface dynamics of the proton to define a screening mechanism of the electromagnetic fluctuations at a given scale. From an initial screening at the reduced Compton wavelength of the proton, we find a direct relation to Einstein field equations and the Schwarzschild solution describing a source term for the internal energy of the proton emerging from zero-point electromagnetic fluctuations. A second screening of the vacuum fluctuations is found at the proton charge radius, which accurately results in the rest mass. Considering the initial screening, we compute the Hawking radiation value of the core Schwarzschild structure and find it to be equivalent to the rest mass energy diffusing in the internal structure of the proton. The resulting pressure gradient or pressure forces are calculated and found to be a very good fit to all the measured values of the color force and residual strong force typically associated to quark-antiquark and gluon flux tubes confinement. As a result, we are able to unify all confining forces with the gravitational force emerging from the curvature of spacetime induced by quantum electromagnetic vacuum fluctuations. Finally, we applied the quantum vacuum energy density screening mechanism to the observable universe and compute the correct critical energy density typically given for the total mass-energy of the universe.

- Video "The Origin of Mass and Nature of Gravity Explained" published in 2024: https://www.youtube.com/watch?v=bxdCY2UPhTk

- Related video "Zero-Point Energy Unifies Physics - Nassim Haramein, DemystifySci #357" (interview and conversation with Nassim Haramein,

published in August 2025): https://www.youtube.com/watch?v=uWvv2YQF0CE&t=470s

Significant paper "Extending Einstein-Rosen's Geometric Vision: Vacuum Fluctuations-Induced Curvature as the Source of Mass, Gravity and Nuclear Confinement" by Nassim Haramein, Olivier Alirol, and Cyprien Guermonprez - published in October 2025 via preprints.org: https://doi.org/10.20944/preprints202509.1835.v2 . Abstract of the paper:

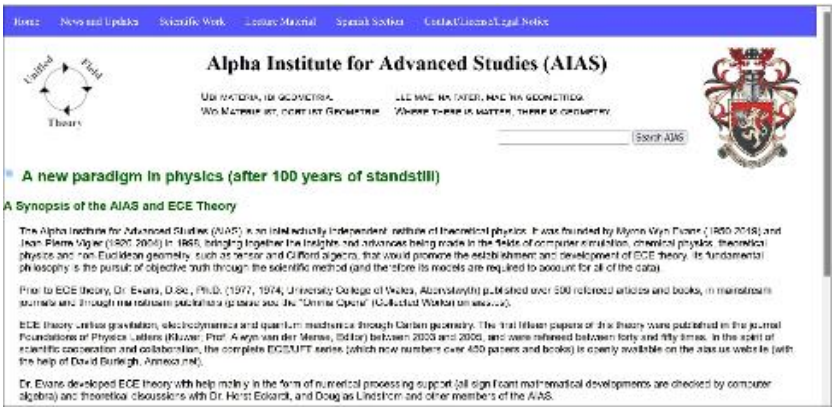
- We derive mass, gravity, and nuclear confinement from first principles through a spacetime mechanism where electromagnetic quantum vacuum fluctuations - originally derived from black-body radiation - induce metric perturbations that serve as a foundational source. Utilizing correlation functions, we demonstrate that highly coherent modes of zero-temperature black-body radiation undergo decoherence within the proton's resonant cavity, producing its exact mass energy density. From Einstein's field equations, we calculate the conversion factor of electromagnetic vacuum fluctuations into gravitational waves traveling through the proton's cavity. We find that this converted energy is equivalent to the energy density of a Kerr-Newman solution at the proton's reduced Compton wavelength, which defines a surface horizon. We compute the Hawking radiation and vaporation of this surface and find it equivalent to the proton's rest mass. The evaporation lifetime far exceeds the observable universe's age and satisfies experimental constraints. We demonstrate that the internal vacuum fluctuations within the proton cavity may provide a stabilizing source term equivalent to Hawking radiation energy, potentially affording extreme stability to the proton. Our formulation reduces to a Klein-Gordon equation on the metric perturbation that yields a confining Yukawa-like energy potential, demonstrating that both confining forces and gravitational forces emerge as consequences of metric perturbations generated by quantum electromagnetic fluctuations. This result aligns well with experimental measurements across multiple scales - from the color force to the residual strong force, and ultimately to gravitation. By computing both confining forces and gravity as emergent manifestations of vacuum fluctuations curving spacetime rather than separate fundamental interactions, we resolve Einstein and Rosen's attempt to geometrize particles and forces at the quantum scale.

The ISF did arise from a previous history comprising 30 years research, publications, and designs as well as the Resonance Science Foundation, education, and a few related smaller companies.

There is a possibility to invest into the ISF / support the ISF via a participatory perpetual bond. The ISF issues in two stages maximal 200 Million notes denominated at 1 CHF per note. The minimum investment is 1000 CHF

Advanced physics and entirely novel and environmentally friendly energy technologies - Alpha Institute for Advanced Studies (AIAS) and ECE Theory

Creation date 3 August 2025



Sreenshot of the home page of the website of the Alpha Institute for Advanced Studies (AIAS), https://aias.institute , from 2 August 2025

Website: https://aias.institute

YouTube channel: https://www.youtube.com/channel/UC1LlhERLMwyV9l1VjnRskIQ

Introduction into the ECE Theory:

- Brief introduction into the ECE Theory: https://aias.institute/1-Home/Brief-Introduction.html . A significant citation from it: "... In ECE theory, there are no sources a priori, only fields. Matter is considered to be a "condensed field" of general relativity, and spacetime itself may be interpreted as a vacuum or aether field that exists everywhere. Matter as condensed fields leads directly to quantum mechanics, and avoids extra concepts like quantum electrodynamics. ..."
- Paper "Einstein, Cartan and Evans – Start of a New Age in Physics?" by Horst Eckardt and Laurence G. Felker, at least version 3 from February 2022: https://aias.institute/documents/eceArticle/ECE-Article_EN.pdf
- Video about the ECE Theory by Horst Eckardt https://www.youtube.com/watch?v=xK9ICtpH99w

Paper "How Do Space Energy Devices Work - An explanation by means of the Einstein-Cartan-Evans (ECE) theory" by Horst Eckardt: https://aias.institute/documents/miscellaneous/SpaceEnergy.pdf

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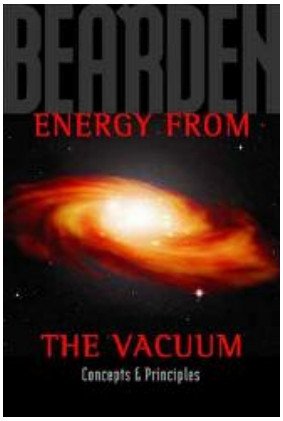
Entirely novel and environmentally friendly energy technologies

Tom Bearden: His website, motionless electromagnetic generator (MEG), and book "Energy from the Vacuum - Concepts and Principles"

Last update 11 August 2024

Tom Bearden passed away on 3 February 2022 and an obituary is published in the following link: https://obits.al.com/us/obituaries/huntsville/name/thomas-bearden-obituary?id=32759244 . A German-language obituary is published on pages 50 - 52 in the March / April 2022 issue of the German-language journal NET-Journal (ISSN 1420-9292)



Book “Energy from the Vacuum – Concepts and Principles”. Published in 2004. Author: Tom Bearden. 977 pages. ISBN 0972514600. Preview Table of Contents

Tom Bearden's former website cheniere.org no longer exists but is still available via the website web.archive.org: https://web.archive.org/web/20220428030850/http://www.cheniere.org . It comprises comprehensive information such as "A Partial List of Successful Documented EM Over-Unity and Negative Resistor Devices and Processes": https://web.archive.org/web/20220429004028/http://www.cheniere.org/misc/oulist.htm

The Motionless Electromagnetic Generator (MEG) of Tom Bearden is presented in https://web.archive.org/web/20220301131053/https://www.cheniere.org/megstatus.htm . In an e-mail to a professional electronics engineer from April 2003 he explains how the MEG works: https://web.archive.org/web/20220301131053/https://www.cheniere.org/correspondence/041303.ht or download it as pdf via this website: https://novam-research.com/resources/MEG_How-it-works_e-mail-from-Tom-Bearden-to-a-professional-electronics-engineer-from-April-2003.pdf

The MEG is also explained in the following paper:

Explanation of the motionless electromagnetic generator with O(3) electrodynamics, P. K. Anastasovski, T. E. Bearden et al., Foundation of Physics Letters 14 (2001) 87 - 94, https://doi.org/10.1023/A:1012085429802

Abstract of this paper:

Recently, Bearden el al. developed a device which is known as a motionless electromagnetic generator (MEG) and which produces a coefficient of performance (COP) far in excess of unity. The device has been independently replicated by Naudin. In this communication, the fundamental operational principle of the MEG is explained using a version of higher symmetry electrodynamics known as O(3) electrodynamics, which is based on the empirical existence of two circular polarization states of electromagnetic radiation, and which has been developed extensively in the literature. The theoretical explanation of the MEG with O(3) electrodynamics is straightforward: Magnetic energy is taken directly ex vacua and used to replenish the permanent magnets of the MEG device, which therefore produces a source of energy that, in theory, can be replenished indefinitely from the vacuum. Such a result is incomprehensible in U(1) Maxwell-Heaviside electrodynamics.

Examples of papers including abstract about physical concepts for entirely novel and environmetally friendly energy and propulsion technologies

Last update 2 February 2026

Classical Electrodynamics without the Lorentz Condition: Extracting Energy from the Vacuum

P. K. Anastasovski, T. E. Bearden et al.

Physica Scripta 61 (2000) 513 – 517

https://doi.org/10.1238/Physica.Regular.061a00513

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Classical Electrodynamics without the Lorentz Condition: Extracting Energy from the Vacuum

Abstract: It is shown that if the Lorentz condition is discarded, the Maxwell–Heaviside field equations become the Lehnert equations, indicating the presence of charge density and current density in the vacuum. The Lehnert equations are a subset of the

O(3) Yang–Mills field equations. Charge and current density in the vacuum are defined straightforwardly in terms of the vector potential and scalar potential, and are conceptually similar to Maxwell's displacement current, which also occurs in the classical vacuum. A demonstration is made of the existence of a time dependent classical vacuum polarization which appears if the Lorentz condition is discarded. Vacuum charge and current appear phenomenologically in the Lehnert equations but fundamentally in the O(3) Yang–Mills theory of classical electrodynamics. The latter also allows for the possibility of the existence of vacuum topological magnetic charge density and topological magnetic current density. Both O(3) and Lehnert equations are superior to the Maxwell–Heaviside equations in being able to describe phenomena not amenable to the latter. In theory, devices can be made to extract the energy associated with vacuum charge and current.

PHYSICAL REVIEW E VOLUME 48, NUMBER 2 AUGUST 1993

Extracting energy and heat from the vacuum

Daniel C. Cole
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Harold E. Puthoff
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Received 22 March 1993

Relatively recent proposals have been made in the literature for extracting energy and heat from electromagnetic zero-point radiation via the use of the Casimir force. The basic thermodynamics involved in these proposals is analyzed and clarified here, with the conclusion that, yes, in principle, these proposals are correct. Technological considerations for actual application and use are not examined here, however.

PACS number(s): 05.90.+m, 03.65.−w, 05.40.+j, 05.70.−a

Nearly a decade ago Forward [1] raised the possibility of extracting energy from the electromagnetic zero-point (ZP) fields that are predicted by quantum electrodynamics to be present in all of space. He described a means of accomplishing this task via a mechanical device consisting of a charged foliated conductor. By using the attractive Casimir force between metal layers to overcome a repulsive electrostatic force, the foliated conductor can be greatly compressed, thereby storing charge at a high electrostatic potential energy.

More recently, one of us (Puthoff) has again raised the possibility of energy extraction from the vacuum [2], while also emphasizing that the energy density of the electromagnetic ZP energy has conservatively been estimated to be equal to or greater than nuclear energy densities [3]. Puthoff suggested a potentially more practical and plausible means for energy extraction, namely, a method involving a charged plasma. His discussion included the idea of generating heat from the vacuum.

Here we do not concurrent further on devising practical methods for enabling the vacuum to become a viable, economical alternative to more conventional sources of energy, except to say that, without a doubt, considerable technological effort might need to be expended to adequately harness such energy. Instead, here we will concentrate on the issue of whether fundamental thermodynamic laws are being violated in even considering this possible source of energy. In particular, certainly the "vacuum" should be considered to be a state of thermal equilibrium at the temperature of $T=0$. How then can energy be extracted, and even heat generated, at $T=0$?

Some relatively recent articles by one of us (Cole) provide a starting point for this discussion. These articles analyze the thermodynamics of quasistatic displacement operations on fluctuating electric dipole harmonic oscillators [4–7] and on conducting parallel plates [8]. The operations involve, respectively, the microscopic van der Waals force between atomic systems, and the macroscopic Casimir force between parallel plates. Due to the fundamental thermodynamic definition of $T=0$, no heat flow should occur at $T=0$ during quasistatic displacements of these systems. Indeed, for these two systems, as treated via classical physics, only one electromagnetic thermal radiation spectrum was found to ensure that no heat would flow: namely, the classical electromagnetic ZP radiation spectrum, which has the same spectral form as the ZP spectrum predicted to exist via QED. The existence of this radiation results in van der Waals and Casimir forces at $T=0$, thereby yielding a tight connection between the required spectrum and the resulting forces.

At first thought, a contradiction appears inevitable between the analysis yielding "no heat flow at $T=0$" and "heat extraction at $T=0$." However, the contradiction becomes resolved upon recognizing that two different types of thermodynamic operations are being discussed. The quasistatic operations are thermodynamically reversible, so here no heat flow occurs at $T=0$. In contrast, the heat generation process discussed in Ref. [2] is thermodynamically irreversible, so heat can be produced, even when the initial temperature is $T=0$.

The following analysis will cover both the $T=0$ and $T\neq 0$ cases. Indeed, although the proposals in Refs. [1,2] discussed only the very idealized $T=0$ case, they can be shown to be valid also at $T\neq 0$. Our analysis will use classical physics arguments, as in Refs. [4–8].

The mechanism for heat generation is illustrated in the following thought experiment, which clearly is an impractical process, yet it embodies the necessary points. Suppose there exists a large number of uncharged parallel plate capacitors. The plates of each capacitor will be attracted to each other by the fluctuating, yet correlated, induced charge distributions in each plate, that arise on account of the fluctuating ZP plus thermal radiation fields. If each pair of plates is allowed to collide, some of the kinetic energy generated will be converted into heat. Collecting the useable portion of the heat, discarding each pair of plates, and then colliding the next set, in turn, then yields a means for heat generation. The "fuel" here is the supply of capacitors; the used up capacitors are analogous to the exhaust from gasoline engines or the "waste" from nuclear fuel.

To analyze this process more deeply, the physical description of colliding systems needs to be addressed. In

1063-651X/93/48(2)/1562(4)/$06.00 48 1562 ©1993 The American Physical Society

Extracting Energy and Heat from the Vacuum

D. C. Cole and H. E. Puthoff

Physical Review E 48 (1993) 1562 - 1565

https://doi.org/10.1103/PhysRevE.48.1562

Published also in the website of the Institute for Advanced Studies at Austin via the following link: http://www.earthtech.org/publications/PREv48_1562.pdf

Abstract: Relatively recent proposals have been made in the literature for extracting energy and heat from electromagnetic zero-point radiation via the use of the Casimir force. The basic thermodynamics involved in these proposals is analyzed and clarified here, with the conclusion that, yes, in principle, these proposals are correct. Technological considerations for actual application and use are not examined here, however.

Explanation of the motionless electromagnetic generator with O(3) electrodynamics

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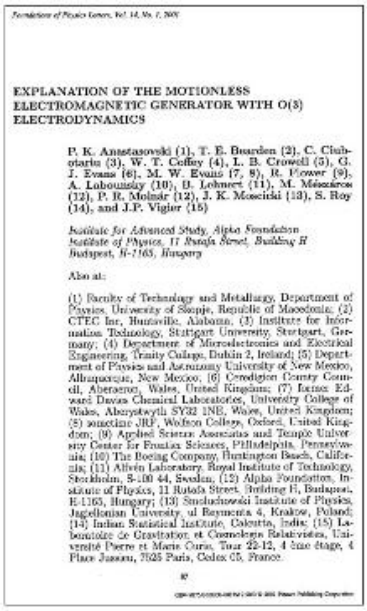

EXPLANATION OF THE MOTIONLESS ELECTROMAGNETIC GENERATOR WITH O(3) ELECTRODYNAMICS

P. K. Anastasovski (1), T. E. Bearden (2), C. Ciubotariu (3), W. T. Coffey (4), L. B. Crowell (5), G. J. Evans (6), M. W. Evans (7, 8), R. Flower (9), A. Labounsky (10), B. Lehnert (11), M. Mészáros (12), P. R. Molnár (12), J. K. Moscicki (13), S. Roy (14), and J.P. Vigier (15)

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Abstract: Recently, Bearden el al. developed a device which is known as a motionless electromagnetic generator (MEG) and which produces a coefficient of performance (COP) far in excess of unity. The device has been independently replicated by Naudin. In this communication, the fundamental operational principle of the MEG is explained using a version of higher symmetry electrodynamics known as O(3) electrodynamics, which is based on the empirical existence of two circular polarization states of electromagnetic radiation, and which has been developed extensively in the literature. The theoretical explanation of the MEG with O(3) electrodynamics is straightforward: Magnetic energy is taken directly ex vacua and used to replenish the permanent magnets of the MEG device, which therefore produces a source of energy that, in theory, can be replenished indefinitely from the vacuum. Such a result is incomprehensible in U(1) Maxwell-Heaviside electrodynamics.

Increased voltage phenomenon in a resonance circuit with unconventional magnetic configuration

Osamu Ide

Journal of Applied Physics 77 (1995) 6015 - 6020

https://doi.org/10.1063/1.359520

Also available via http://www.intalek.com/Index/Projects/Research/jap77.pdf

Abstract: The behavior of an LCR (inductance-capacitance-resistance) circuit with a movable ferromagnetic core is discussed. The core is attracted by a magnetic field generated by an electric current resulting from the discharge of a capacitor in the closed LCR circuit. An unusual increase in recharge voltage, which was dependent on the magnetic configuration of the coil, was observed. This voltage increase does not conform to the mathematical simulation of the system. The possibility that a positive electromotive force was involved in this effect is discussed.

A paper about the so-called EM Drive:

Measurement of Impulsive Thrust from a Closed Radio-Frequency Cavity in Vacuum

H. White et al.

Journal of Propulsion and Power 33 (2017) 830 - 841

https://doi.org/10.2514/1.B36120

Instead of the abstract we present here a significant citation from page 840: "... This suggests that the idea of treating the quantum vacuum as a dynamic medium capable

of supporting oscillations might be valid. If a medium is capable of supporting acoustic oscillations, this means that the internal constituents were capable of interacting and exchanging momentum. If the vacuum is indeed mutable and degradable as was explored, then it might be possible to do / extract work on / from the vacuum, and thereby be possible to push off of the quantum vacuum and preserve the laws of conservation of energy and conservation of momentum ..."

A review on how a Perpetual Motion Machine generates electrical power

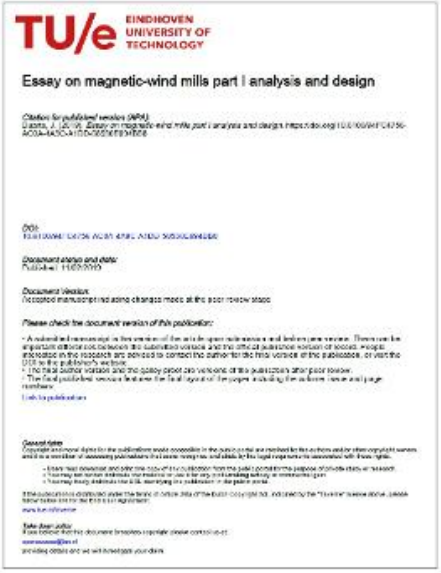
M. N. Hidayat, S. P. Chairandy, and F. Ronilaya

IOP Conference Series: Materials Science and Engineering 1098 (2021) 042063 (1 - 7)

https://doi.org/10.1088/1757-899X/1098/4/042063

Abstract: Electrical energy demand is increasing in accordance with rapid growth of the human population. Since fossil fuels is the most widely used energy source, thus it is depleting very fast. Alternative energy source is urgently needed to replace the use of conventional energy sources. Perpetual Motion Machine (PMM) which can be applied to produce electricity, may be an alternative solution for the problem the world is facing today. The machine is designed to generate power from repulsive forces of permanent magnet without utilizing external sources. Some researches had conducted experiments and Neodymium magnet is most used in the project due to its strong magnetic field. The device is mainly built using a permanent magnet, a rotating wheel and a generator. This paper reviews some aspects on how A Perpetual Motion Machine (PMM) generates electrical power. The aim of the paper is to provide a summary of the topic and its opportunities in further enhancements for better results. The study found that the concept is very effective, ecofriendly and less space needed. However, a larger scale development of the machine along with proper magnet and gear arrangement is currently needed for a better performance and application.

Paper about a discussion of the calculation of magnetic fields of permanent magnets and arrangements of permanent magnets which lead to a self-running motion


TU/e EINDHOVEN UNIVERSITY OF TECHNOLOGY

Essay on magnetic-wind mills part I analysis and design

Essay on Magnetic-Wind Mills - Part 1: Analysis and Design

J. L. Duarte (2019)

https://doi.org/10.6100/94FC4756-AC0A-4A9C-A1DD-58530E694BB8

Abstract: A methodology for the synthesis of a prime mover is presented, based only on attraction / repulsion of permanent magnets. The design example is given for a demonstration prototype that has the potential to unfold, in theory, the sustainable generation of 22 W mechanical power at 1000 rpm and beyond.

A version from Part 2, namely that from 15 November 2022, can be downloaded from this website: https://novam-research.com/resources/Essay-on-Magnetic-Wind-Mills_Part2_J-L-Duarte_v4_15-Nov-2022.pdf . Abstract: One of the missing links in support of the allegation that devices with only magnets seemingly generate useful energy ”out of thin air” could be Spacetime twisting. This discussion paper applies the concept to self-rotating machines previously claimed in patents.

Kufa Journal of Engineering
Vol. 6, No. 2, June, P.P.24-38
Received 3 June 2014, accepted 28 December 2014

DESIGN OF OPEN SOURCE STRAIGHT PERMANENT MAGNET MOTOR

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2 Mid. Euphrates Branch, The general company of Electricity Prod. haider-81@live.com

ABSTRACT

This paper deals with the attraction and repulsion properties of the magnetic field of permanent magnets in order to produce reciprocating motion in a member of magnetizing material.

The essential idea of this work have started by putting a string of eight magnets type (Nd Fe 35) spaced by equal spaces placed on the axis between two series of same magnet type each string consists of nine magnets pieces spaced apart vacuum. Series of internal and external were organized with different opposite polarity north and south that created a series of free movement of internal series between the two external strings.

First of all, designed magnetic motor rectal movement consisting of two fixed (for external strings) and one moving part (Series Interior) analysis was done and its operation was analyzed. In addition, equations that describe the magnetic field arising from the moving part were modified by using two-dimensional computer model. The (femm) program was used for modeling the distribution of density, and DS Solid Works program to study and calculate the magnetic flux density, field strength and torque generated.

KEYWORDS

Attraction, Flux Density, Free Open Source, Permanent Magnet Motor and Repulsion

This paper presents in its introduction various types of magnet motors and reports about an own concept

Design of open source straight permanent magnet motor

Amel Ridha and Haider H. Jabber

Kufa Journal of Engineering, Volume 6 (2015), pages 24 - 38

https://doi.org/10.30572/2018/KJE/621146

Abstract: This paper deals with the attraction and repulsion properties of the magnetic field of permanent magnets in order to produce reciprocating motion in a member of magnetizing material. The essential idea of this work have started by putting a string of eight magnets type (Nd Fe 35) spaced by equal spaces placed on the axis between two series of same magnet type each string consists of nine magnets pieces spaced apart vacuum. Series of internal and external were organized with different opposite polarity north and south that created a series of free movement of internal series between the two external strings. First of all, designed magnetic motor rectal movement consisting of two fixed (for external strings) and one moving part (Series Interior) analysis was done and its operation was analyzed. In addition, equations that describe the magnetic field arising from the moving part were modified by using two-dimensional computer model. The (femm) program was used for modeling the distribution of density, and DS Solid Works program to study and calculate the magnetic flux density, field strength and torque generated.

Significant paper about the physics of the (quantum) vacuum:

The Origin of Mass and the Nature of Gravity

Nassim Haramein, Cyprien Guermonprez, and Olivier Alirol

Published on 26 September 2023 via zenodo.org:
https://doi.org/10.5281/zenodo.10125315

The Origin of Mass and the Nature of Gravity

Abstract

Introduction

Abstract: From the early explorations of thermodynamics and characterization of black body radiation, Max Planck predicted the existence of a non-zero expectation value for the electromagnetic quantum vacuum energy density or zero-point energy (ZPE). From the mechanics of a quantum oscillator, Planck derived the black body spectrum, which satisfied the Stefan-Boltzmann law with a non-vanishing term remaining where the summation of all modes of oscillations diverged to infinity in each point of the field. In modern derivation, correlation functions are utilized to derive the coherent behavior of the creation and annihilation operators. Although a common approach is to normalize the Hamiltonian so that all ground state modes cancel out, setting artificially ZPE to zero, zero-point energy is essential for the mathematical consistency of quantum mechanics as it maintains the non-commutativity of the creation and annihilation operators resulting in the Heisenberg uncertainty principle. From our computation, we demonstrate that coherent modes of the correlation functions at the characteristic time of the proton correctly result in the emergence of its mass directly from quantum vacuum fluctuation modes. We find as well that this energy value is consistent with a Casimir cavity of the same characteristic distance. As a result, we developed an analytical solution describing both the structure of quantum spacetime as vacuum fluctuations and extrapolate this structure to the surface dynamics of the proton to define a screening mechanism of the electromagnetic fluctuations at a given scale. From an initial screening at the reduced Compton wavelength of the proton, we find a direct relation to Einstein field equations and the Schwarzschild solution describing a source term for the internal energy of the proton emerging from zero-point electromagnetic fluctuations. A second screening of the vacuum fluctuations is found at the proton charge radius, which accurately results in the rest mass. Considering the initial screening, we compute the Hawking radiation value of the core

Schwarzschild structure and find it to be equivalent to the rest mass energy diffusing in the internal structure of the proton. The resulting pressure gradient or pressure forces are calculated and found to be a very good fit to all the measured values of the color force and residual strong force typically associated to quark-antiquark and gluon flux tubes confinement. As a result, we are able to unify all confining forces with the gravitational force emerging from the curvature of spacetime induced by quantum electromagnetic vacuum fluctuations. Finally, we applied the quantum vacuum energy density screening mechanism to the observable universe and compute the correct critical energy density typically given for the total mass-energy of the universe.

The following paper is about the so-called ECAT technology which was / is considered as a LENR-based technology. LENR means Low Energy Nuclear Reactions which usually imply no harmful and radioactive byproducts. Broadly speaking it was assumed that hydrogen H (1) and nickel Ni (28) merge via LENR to copper Cu (29) which releases a large amount of energy. Meanwhile Andrea Rossi considers the underlying physics in terms of dense electron clusters, electron-proton aggregates at picometer scale, and their formation via vacuum polarization and zero point energy.

E-Cat SK and long-range particle interactions

Andrea Rossi (2019) (last update 22 October 2022)

https://dx.doi.org/10.13140/RG.2.2.28382.

Or click here (1) to download it from ecatthenewfire.com or click here (2) to download it from this website

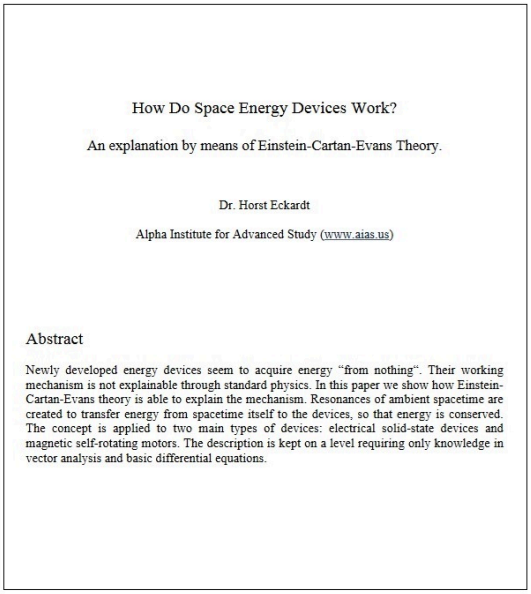
Abstract: Some theoretical frameworks that explore the possible formation of dense exotic electron clusters in the E-Cat SK are presented. Some considerations on the probable role of Casimir, Aharonov-Bohm, and collective effects in the formation of such structures are proposed. A relativistic interaction Lagrangian, based on a pure electromagnetic electron model, that suggests the possible existence of very low

entropy charge aggregates and that highlights the primary role of the electromagnetic potentials in these clusters is presented. The formation of these cluster may be associated to a localized Vacuum polarization generated by a rapid radial charge displacement. The formation of these dense electron clusters are introduced as a probable precursor for the formation of proton-electron aggregates at pico-metric scale, stressing the importance of evaluating the plausibility of special electron-nucleon interactions, as already suggested in [#GullstromRossi]. An observed isotopic dependence of a particular spectral line in the visible range of E-Cat plasma spectrum seems to confirm the presence of a specific proton-electron interaction at electron Compton wavelength scale.

How Do Space Energy Devices Work - An explanation by means of the Einstein-Cartan-Evans (ECE) theory

Horst Eckardt

Published via the AIAS website: https://aias.institute/docum



How Do Space Energy Devices Work?

An explanation by means of Einstein-Cartan-Evans Theory.

Dr. Horst Eckardt

Alpha Institute for Advanced Study (www.aias.us)

Abstract

Newly developed energy devices seem to acquire energy "from nothing". Their working mechanism is not explainable through standard physics. In this paper we show how Einstein-Cartan-Evans theory is able to explain the mechanism. Resonances of ambient spacetime are created to transfer energy from spacetime itself to the devices, so that energy is conserved. The concept is applied to two main types of devices: electrical solid-state devices and magnetic self-rotating motors. The description is kept on a level requiring only knowledge in vector analysis and basic differential equations.

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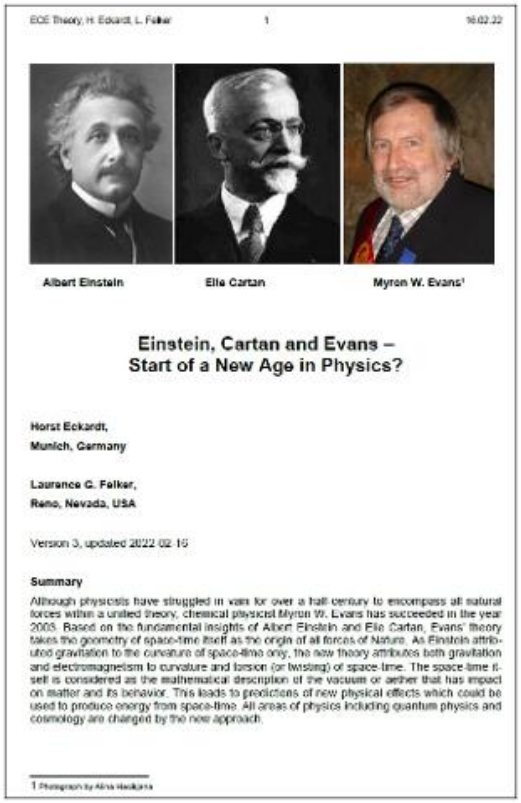
This paper presents an introduction into the Einstein-Cartan-Evans (ECE) theory

Einstein, Cartan and Evans – Start of a New Age in Physics?

Horst Eckardt and Laurence G. Felker

At least version 3 from February 2022

Published via the AIAS website: https://aias.institute/documents/eceArticle/EArticle_EN.pdf



ECE Theory, H. Eckardt, L. Felker 1 16.02.22

Albert Einstein Elie Cartan Myron W. Evans[1]

Einstein, Cartan and Evans – Start of a New Age in Physics?

Horst Eckardt,
Munich, Germany

Laurence G. Felker,
Reno, Nevada, USA

Version 3, updated 2022-02-16

Summary

Although physicists have struggled in vain for over a half-century to encompass all natural forces within a unified theory, chemical physicist Myron W. Evans has succeeded in the year 2003. Based on the fundamental insights of Albert Einstein and Elie Cartan, Evans' theory takes the geometry of space-time itself as the origin of all forces of Nature. As Einstein attributed gravitation to the curvature of space-time only, the new theory attributes both gravitation and electromagnetism to curvature and torsion (or twisting) of space-time. The space-time itself is considered as the mathematical description of the vacuum or aether that has impact on matter and its behavior. This leads to predictions of new physical effects which could be used to produce energy from space-time. All areas of physics including quantum physics and cosmology are changed by the new approach.

1 Photograph by Alina Hankjarva

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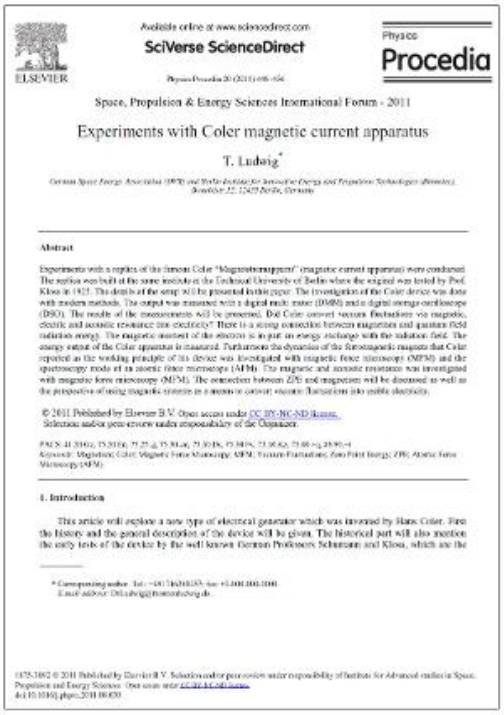

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Physics Procedia

Space, Propulsion & Energy Sciences International Forum - 2011

Experiments with Coler magnetic current apparatus

T. Ludwig*

Abstract

1. Introduction

Experiments with Coler magnetic current apparatus

Thorsten Ludwig

Physics Procedia 20 (2011) 446 - 456

https://doi.org/10.1016/j.phpro.2011.08.039

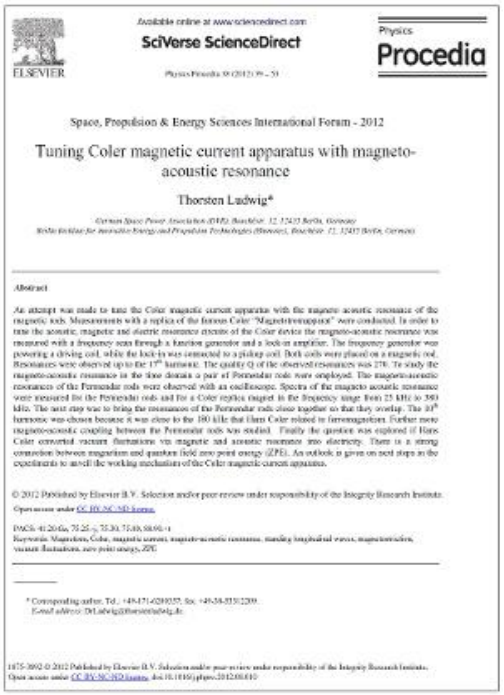
Abstract: Experiments with a replica of the famous Coler “Magnetstromapparat” (magnetic current apparatus) were conducted. The replica was built at the same institute at the Technical University of Berlin where the original was tested by Prof. Kloss in 1925. The details of the setup will be presented in this paper. The investigation of the Coler device was done with modern methods. The output was measured with a digital multi meter (DMM) and a digital storage oscilloscope (DSO). The results of the measurements will be presented. Did Coler convert vacuum fluctuations via magnetic, electric and acoustic resonance into electricity? There is a strong connection between magnetism and quantum field radiation energy. The magnetic moment of the electron is in part an energy exchange with the radiation field. The energy output of the Coler apparatus is measured. Furthermore the dynamics of the ferromagnetic magnets that Coler reported as the working principle of his device was investigated with magnetic force microscopy (MFM) and the spectroscopy mode of an atomic force microscope (AFM). The magnetic and acoustic resonance was investigated with magnetic force microscopy (MFM). The connection between ZPE and magnetism will be discussed as well as the perspective of using magnetic systems as a means to convert vacuum fluctuations into usable electricity.

Tuning Coler Magnetic Current Apparatus with Magneto-Acoustic Resonance

Thorsten Ludwig:

Physics Procedia 38 (2012) 39 - 53

https://doi.org/10.1016/j.phpro.2012.08.010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Physics Procedia

Space, Propulsion & Energy Sciences International Forum - 2012

Tuning Coler magnetic current apparatus with magneto-acoustic resonance

Thorsten Ludwig*

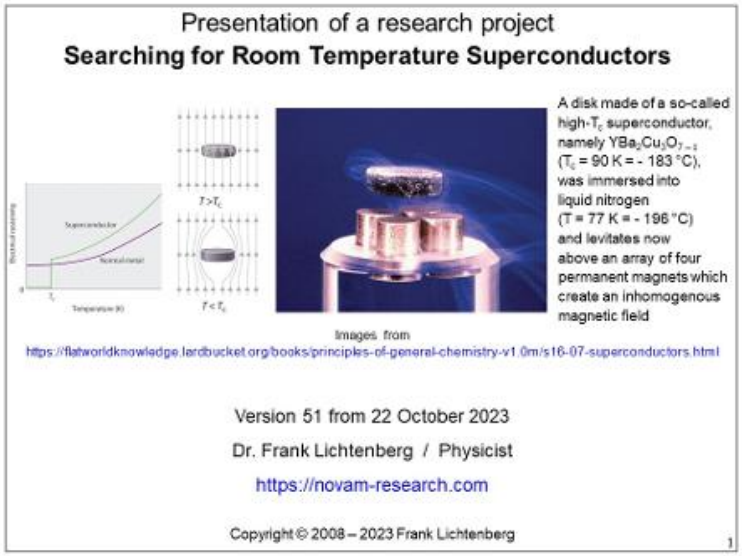
Abstract

Abstract: An attempt was made to tune the Coler magnetic current apparatus with the magneto acoustic resonance of the magnetic rods. Measurements with a replica of the famous Coler “Magnetstromapparat” were conducted. In order to tune the acoustic, magnetic and electric resonance circuits of the Coler device the magneto-acoustic resonance was measured with a frequency scan through a function generator and a lock-in amplifier. The frequency generator was powering a driving coil, while the lock-in was connected to a pickup coil. Both coils were placed on a magnetic rod. Resonances were observed up to the 17th harmonic. The quality Q of the observed resonances was 270. To study the magneto-acoustic resonance in the time domain a pair of Permendur rods were employed. The magneto-acoustic resonances of the Permendur rods were observed with an oscilloscope. Spectra of the magneto acoustic resonance were measured for the Permendur rods and for a Coler replica magnet in the frequency range from 25 kHz to 380 kHz. The next step was to bring the resonances of the Permendur rods close together so that they overlap. The 10thharmonic was chosen because it was close to the 180 kHz that Hans Coler related to ferromagnetism. Further more magneto-acoustic coupling between the Permendur rods was studied. Finally the question was explored if Hans Coler converted vacuum fluctuations via magnetic and acoustic resonance into electricity. There is a strong connection between magnetism and quantum field zero point energy (ZPE). An outlook is given on next steps in the experiments to unveil the working mechanism of the Coler magnetic current apparatus.

Superconductivity

Last update 22 October 2023

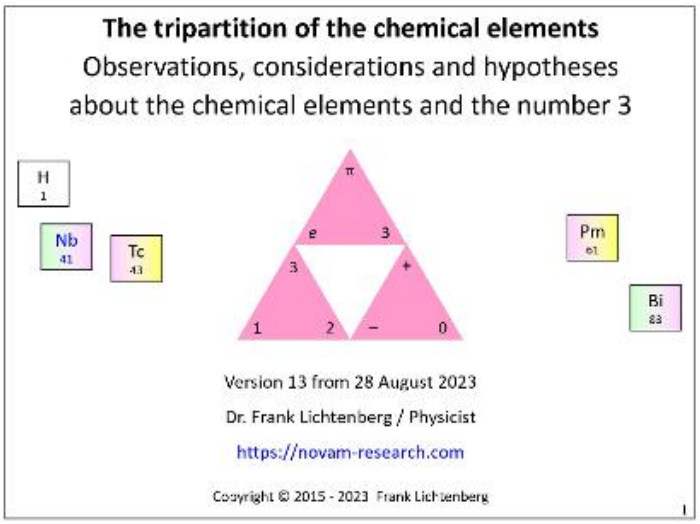
Presentation of a research project: Searching for Room Temperature Superconductors (186 pages / slides, 11 MB pdf, version 51 from 22 October 2023)



The tripartition of the chemical elements

Last update 3 September 2023

Observations, considerations and hypotheses about the chemical elements and the number 3: The tripartition of the chemical elements (89 pages / slides, 1 MB pdf, version 13 from 28 August 2023)



Presentation of a laboratory for the synthesis and study of melt-grown oxides and related topics

11 February 2017

Presentation about a laboratory for the synthesis and study of (melt-grown crystalline) oxides at the ETH Zurich and related topics. This publication from 2017 comprises 438 pages / slides and is published by the library of the ETH Zurich: https://dx.doi.org/10.3929/ethz-a-010817148 . This link allows the download as pdf document or as ppsx type Power Point Show with embedded videos

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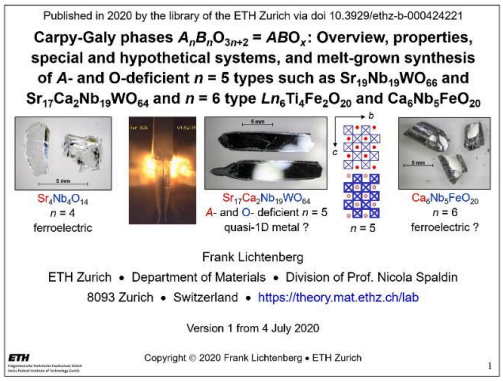
Oxides of the type AnBnO3n+2

9 July 2020

Carpy-Galy phases AnBnO3n+2 = ABOx : Overview, properties, special and hypothetical systems, and melt-grown synthesis of A- and O-deficient n = 5 types such as Sr19Nb19WO66 and Sr17Ca2Nb19WO64 and n = 6 type Ln6Ti4Fe2O20 and Ca6Nb5FeO20. This publication from 2020 comprises 477 pages / slides and is published by the library of the ETH Zurich: https://dx.doi.org/10.3929/ethz-b-000424221 . This link allows the download as pdf document or as ppsx type Power Point Show with 17 embedded videos



Harmful effects of electrosmog, wireless communication technologies, and 5G

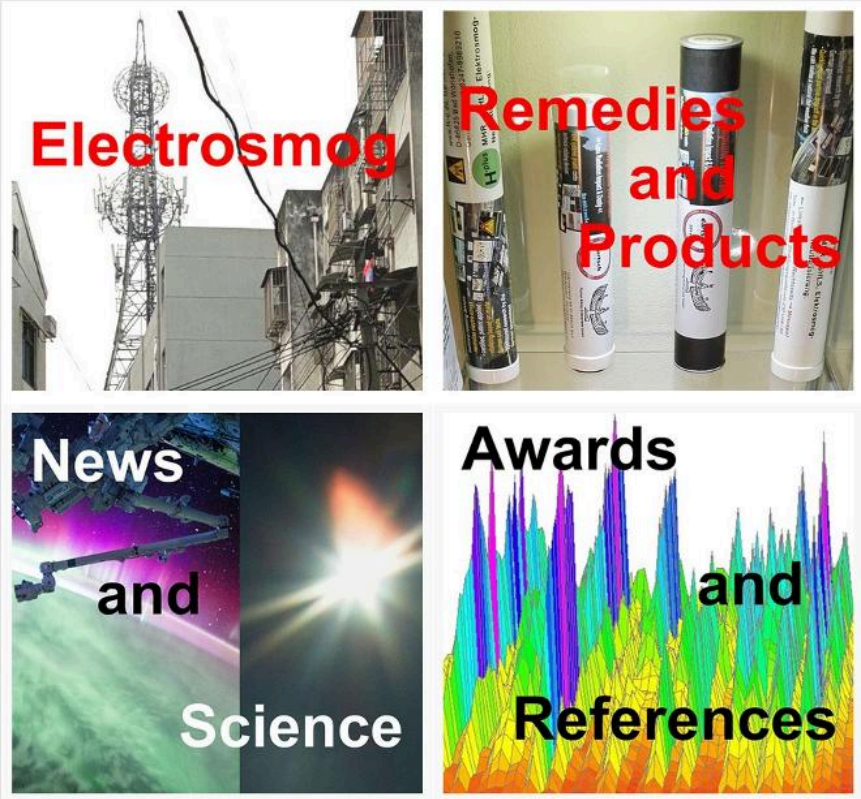
Last update 10 January 2026

German-language website of the Swiss soceity for the

protection against radiation (Verein Schutz vor Strahlung): https://schutz-vor-strahlung.ch

Source of the images on the right: Pictures and logo copied on 10 January 2026 from https://schutz-vor-strahlung.ch

Schutz-vor-Strahlung.ch



Website from Dr. Florian König and his company Florian König Enterprises GmbH - also available in English: https://www.fk-e.de

Image source: English version of https://www.fk-e.de

International appeal for a stop of 5G on earth and in space: https://www.5gspaceappeal.org

Image from https://www.5gspaceappeal.org

Image source and further information: https://ehtrust.org/in-historic-decision-federal-court-finds-fcc-failed-to-explain-why-it-ignored-scientific-evidence-showing-harm-from-wireless-radiation

German-language book "Die Smartphone-Epidemie" (The smart phone epidemic) by Prof. Dr. Dr. Manfred Spitzer, ISBN 978-3-608-98560-3, second printrun from 2020 (in German): https://www.klett-cotta.de/buch/Leben/Die_Smartphone-Epidemie/105963

International Public Symposium "Biological effects of wireless technology" from 4 to 6 October 2019 in Mainz in Germany: https://ehtrust.org/mainz-international-symposium-on-biological-effects-of-wireless

Image from videos which are posted in https://ehtrust.org/mainz-international-symposium-on-biological-effects-of-wireless

Further information, links and websites about harmful effects of wireless communication technologies, 5G, electrosmog, and electromagnetic pollution: https://novam-research.com/electrosmog.php

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